



BLM2502

Theory of

Computation



Pushdown Automata

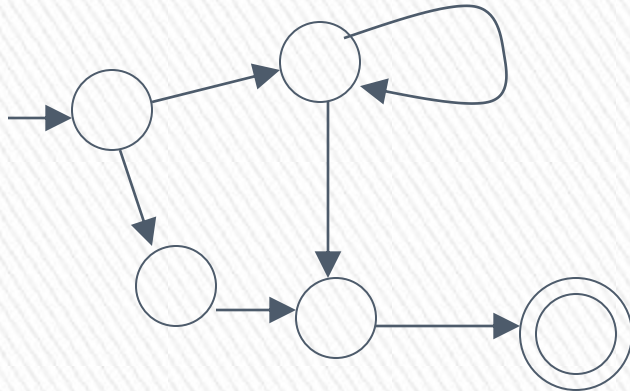
PDA

Pushdown Automaton -- PDA

Input String



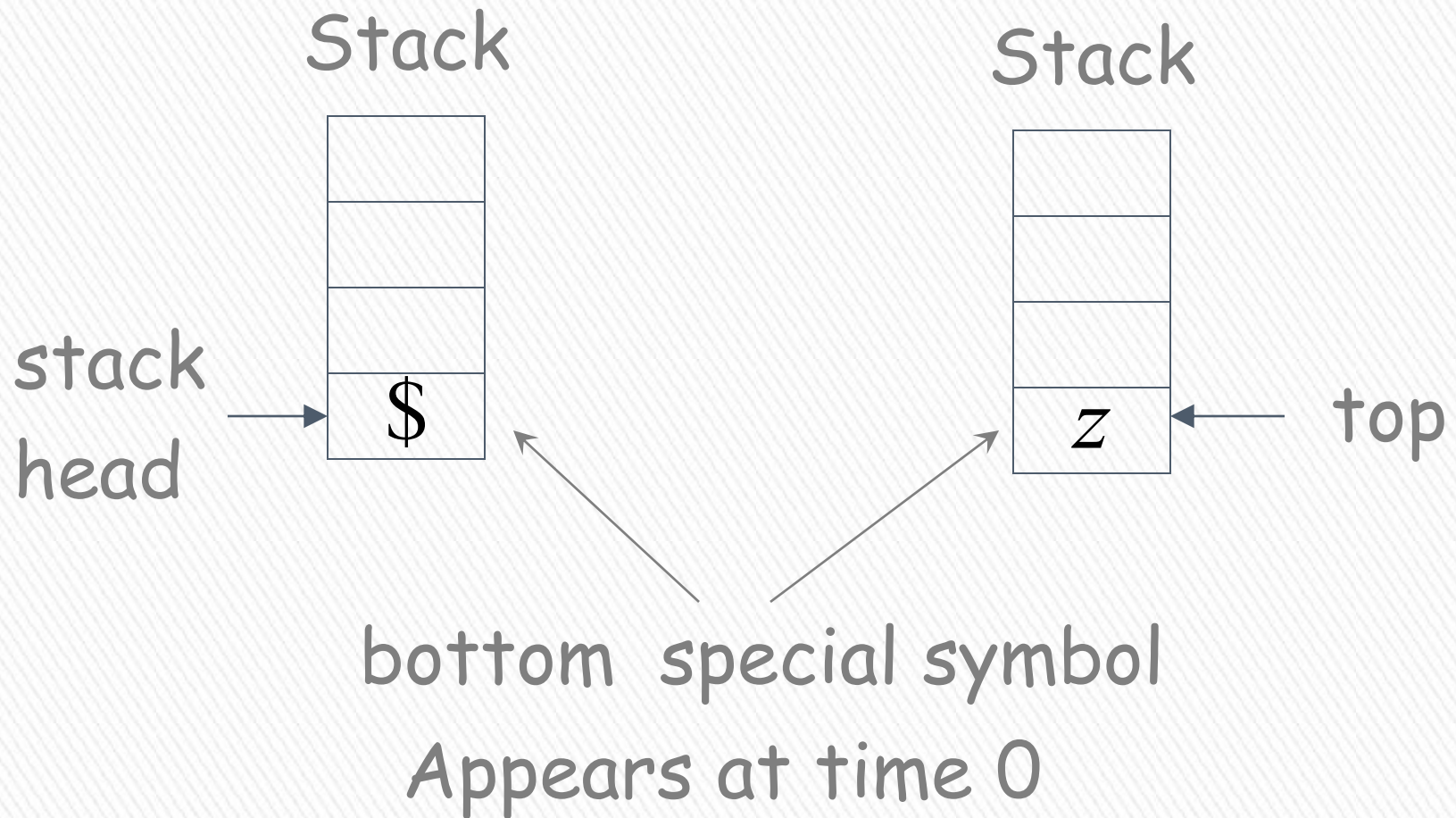
States



Stack



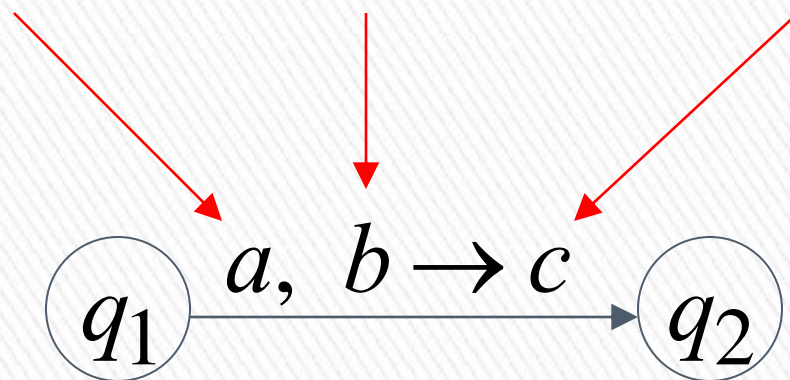
Initial Stack Symbol



Input
symbol

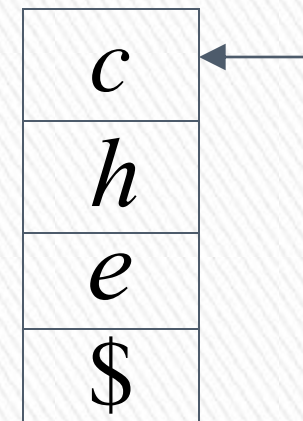
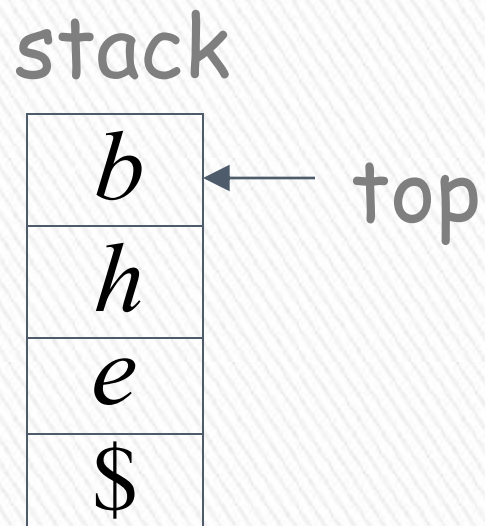
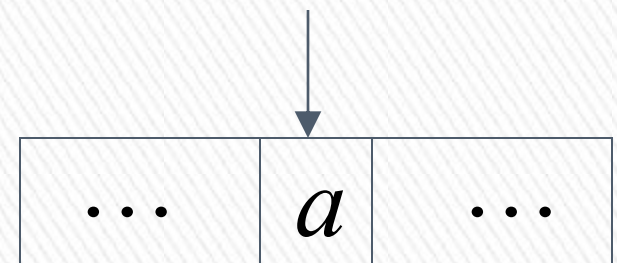
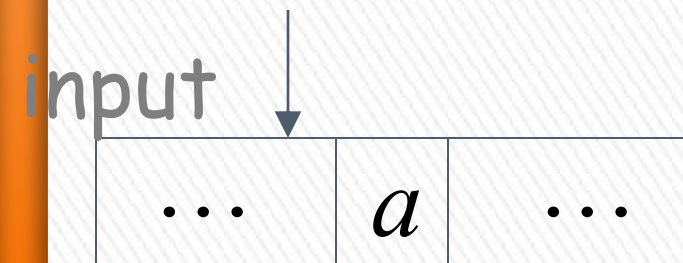
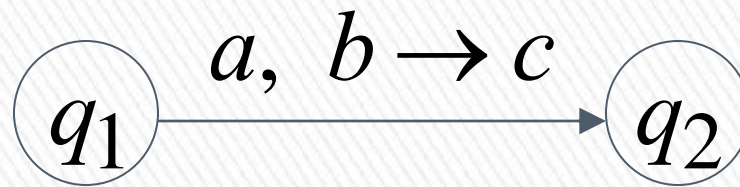
Pop
symbol

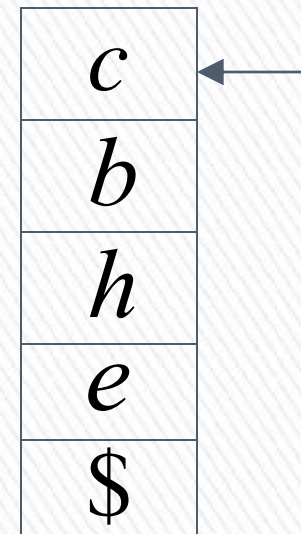
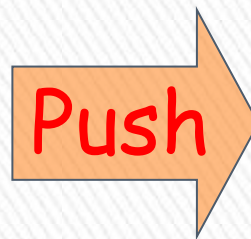
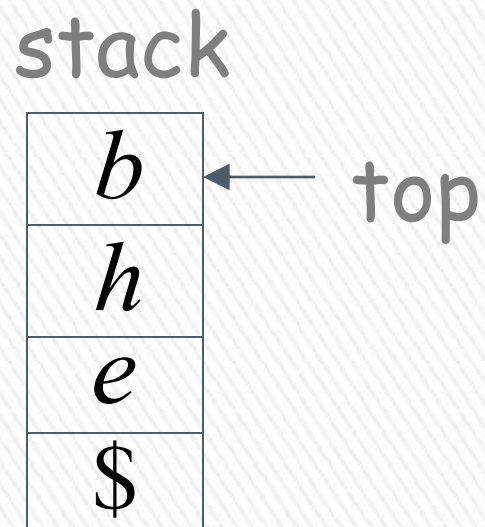
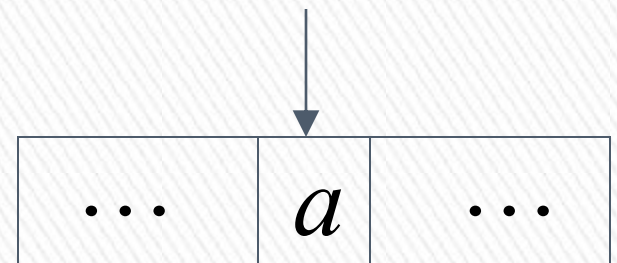
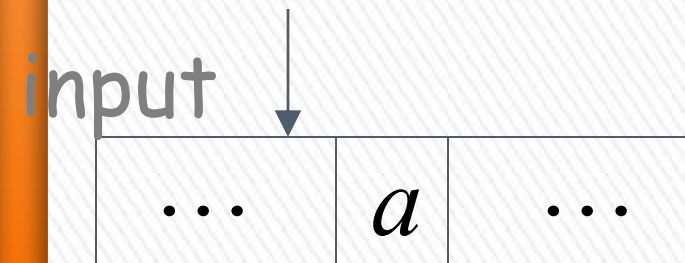
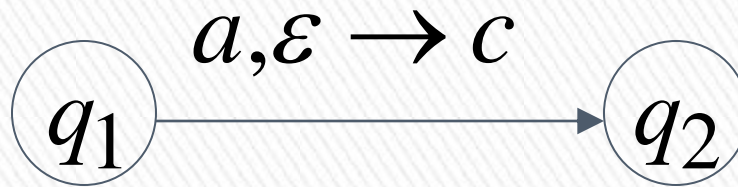
Push
symbol

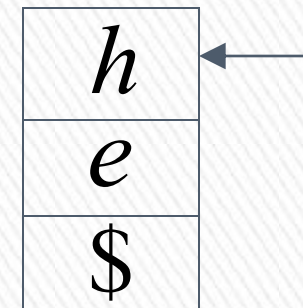
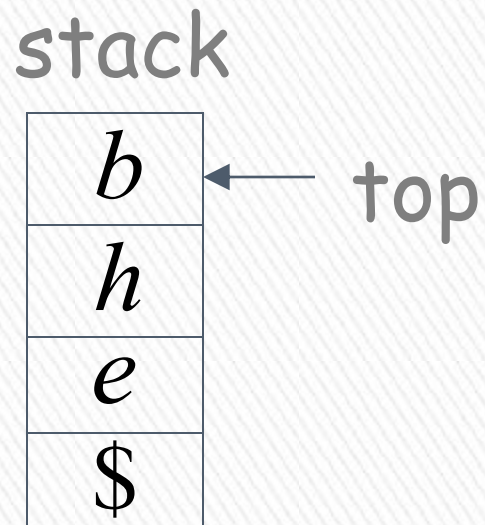
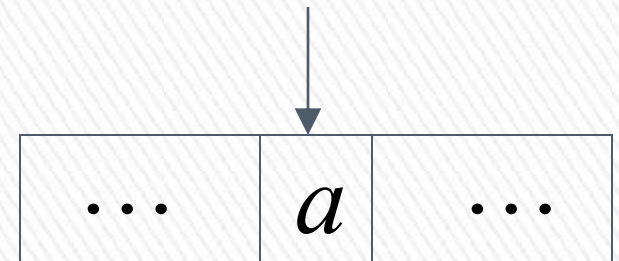
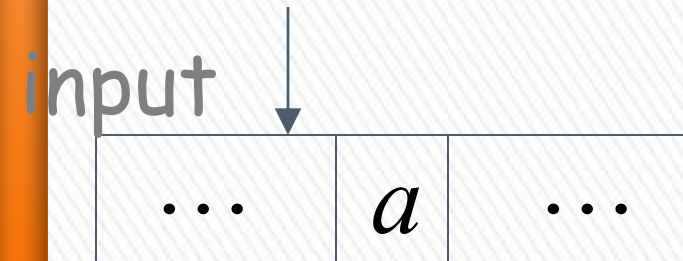
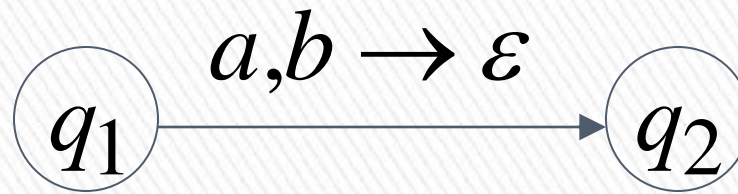


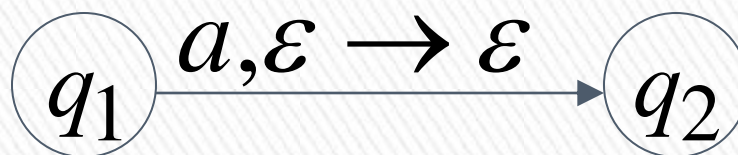
The States



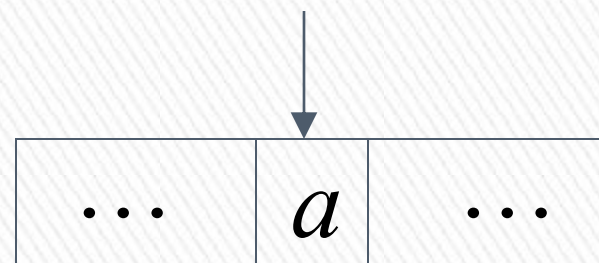
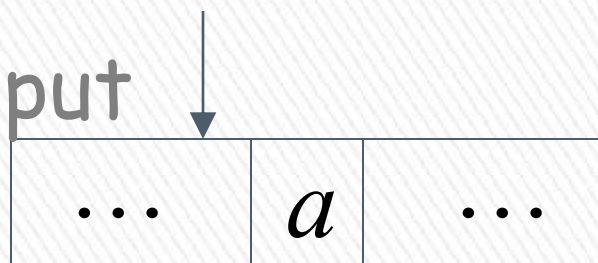




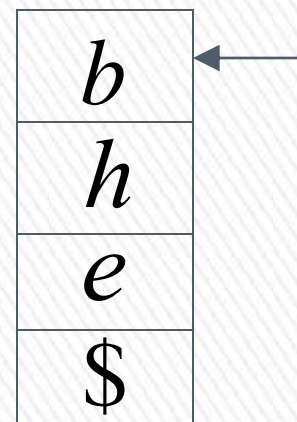
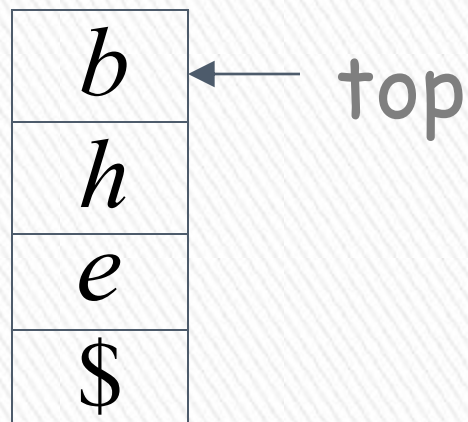




input

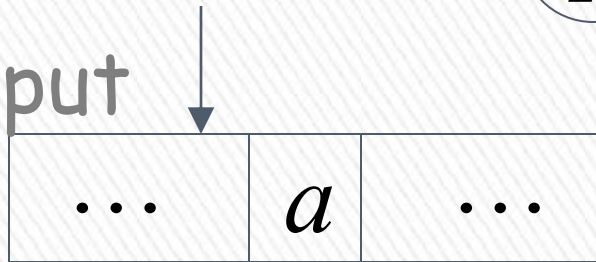


stack

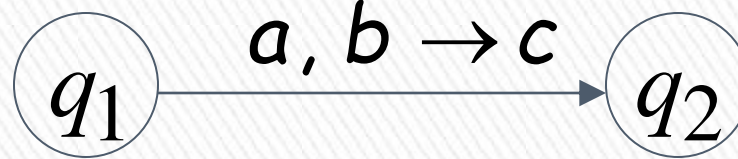


Pop from Empty Stack

input



stack



Stack
underflow

Pop

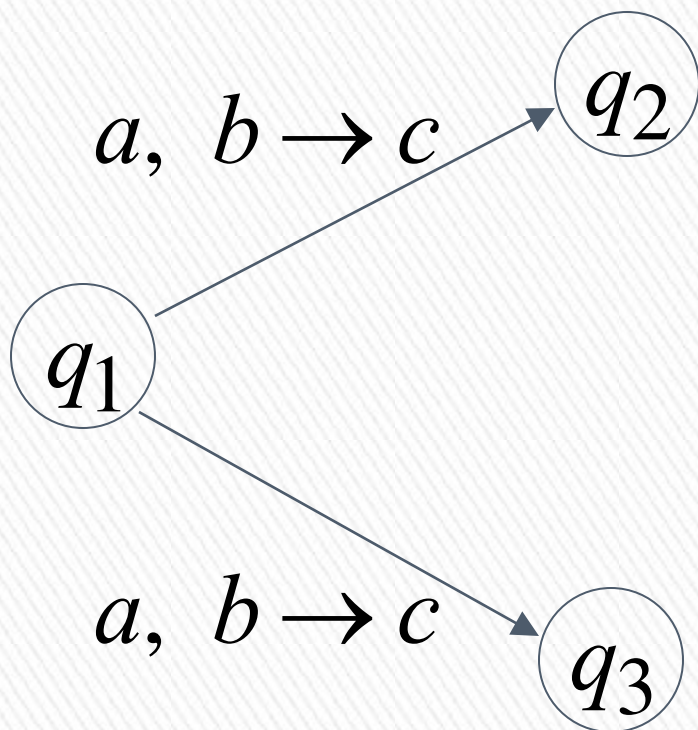
Automaton halts!

If the automaton attempts to pop from empty stack then it halts and rejects input

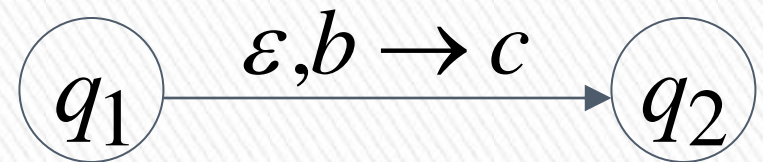
Non-Determinism

PDAs are non-deterministic

Allowed non-deterministic transitions



↳ NFA
§.6:



ϵ – transition

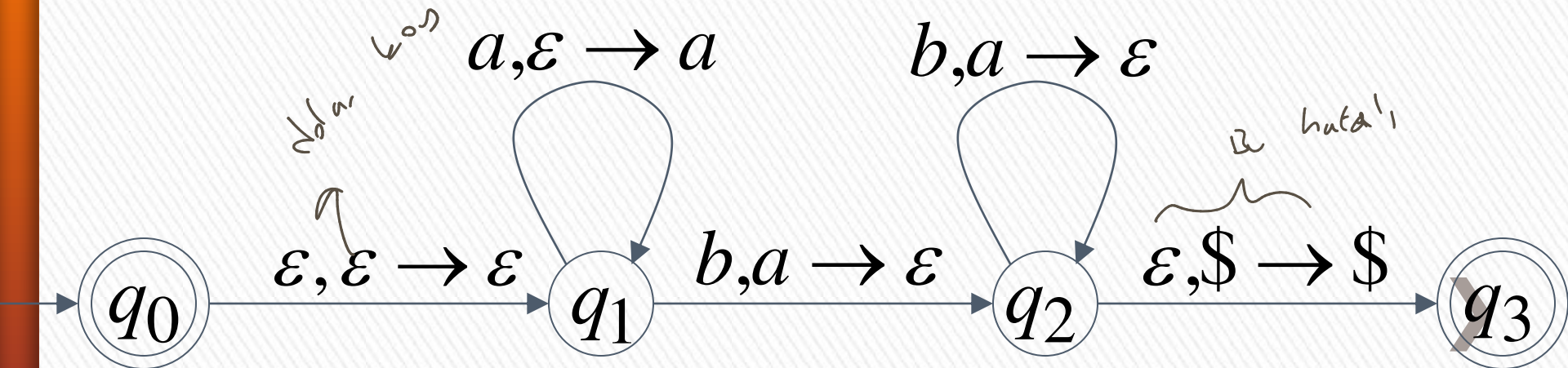


Example PDA

PDA M : $L(M) = \{a^n b^n : n \geq 0\}$

a

b



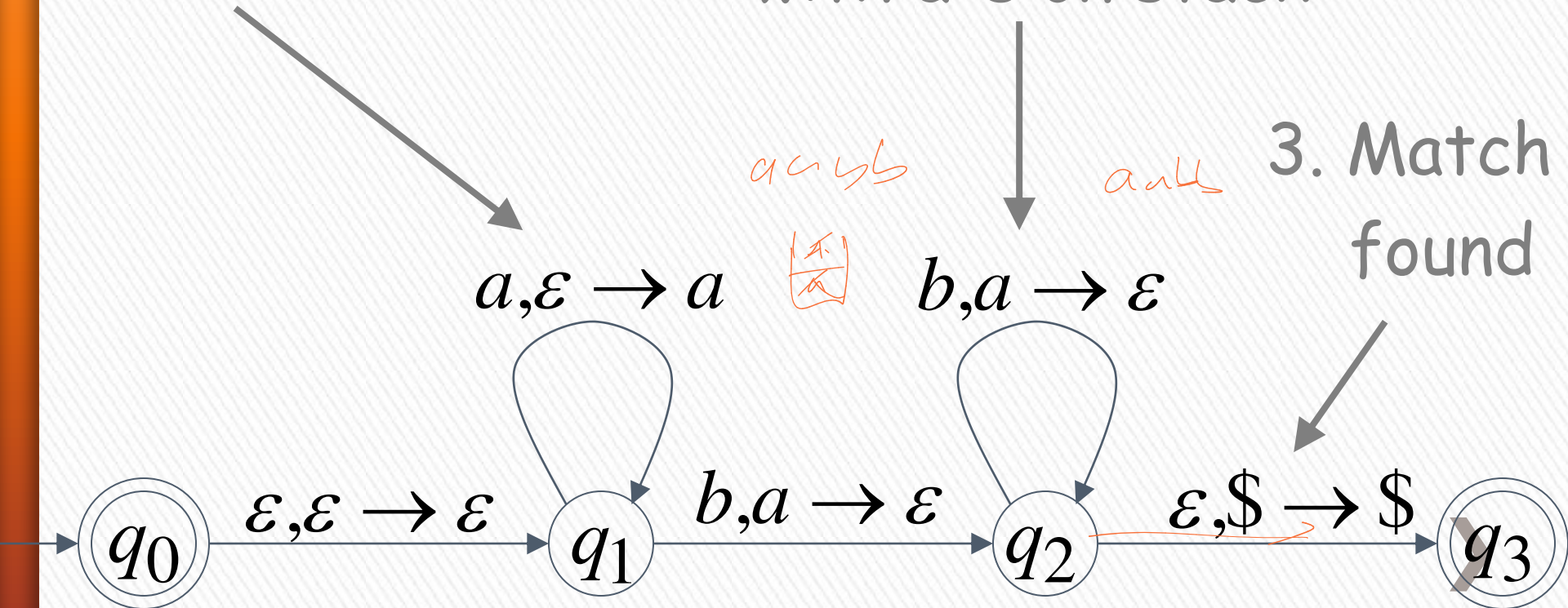
$$L(M) = \{a^n b^n : n \geq 0\}$$

Basic Idea:

1. Push the a's on the stack

2. Match the b's on input with a's on stack

3. Match found



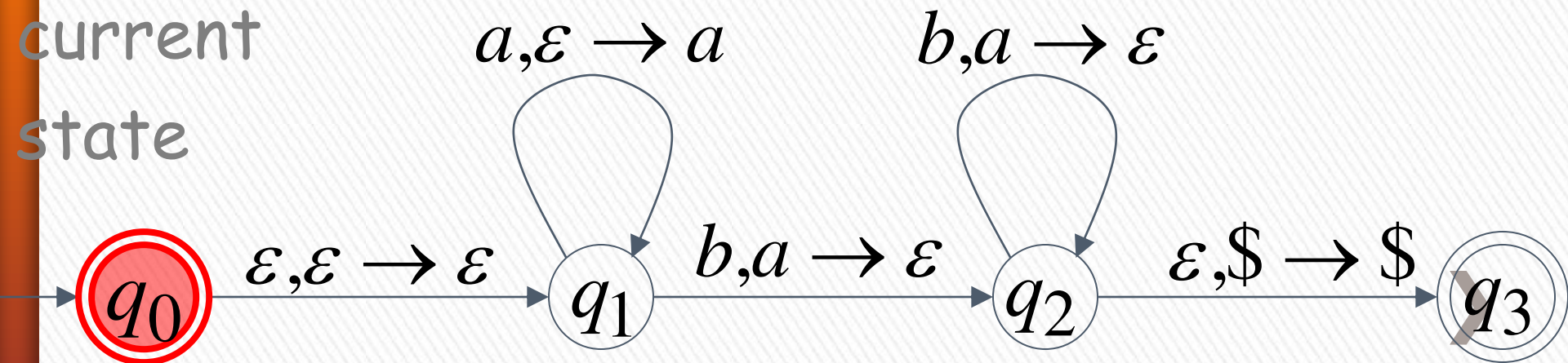
Execution Example: Time 0

Input



Stack

current
state

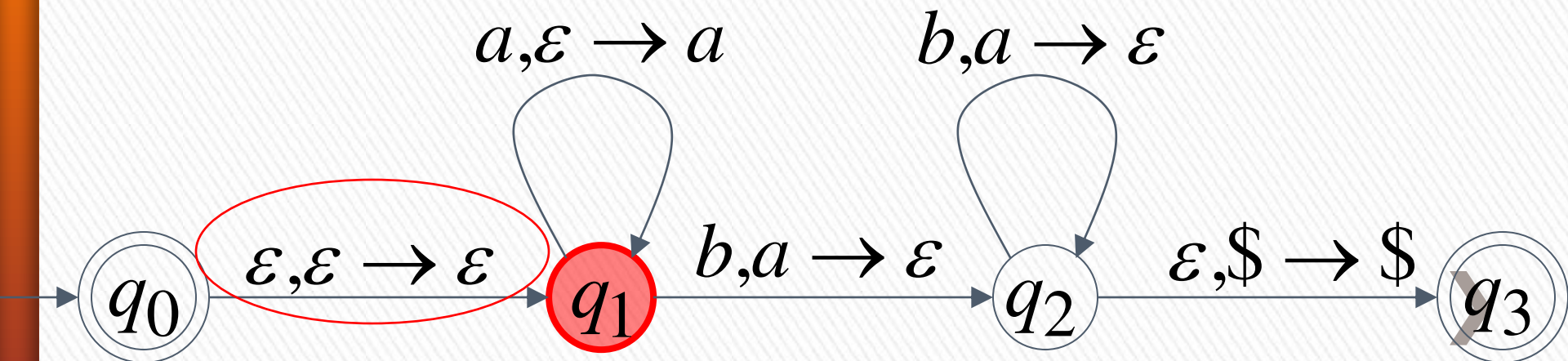


Time 1

Input

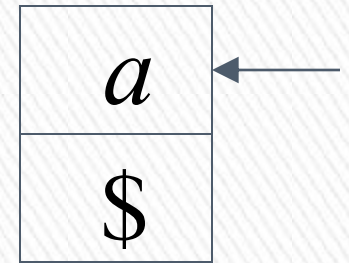
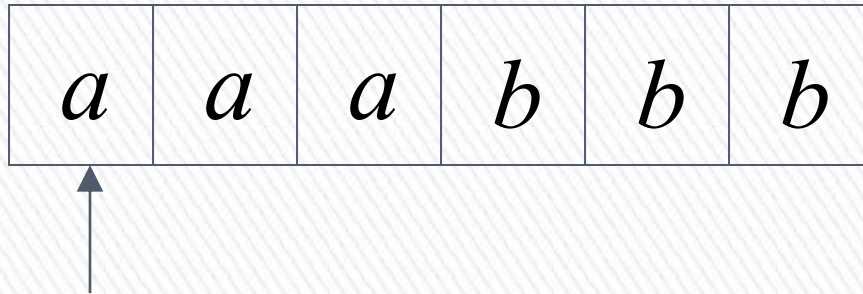


Stack

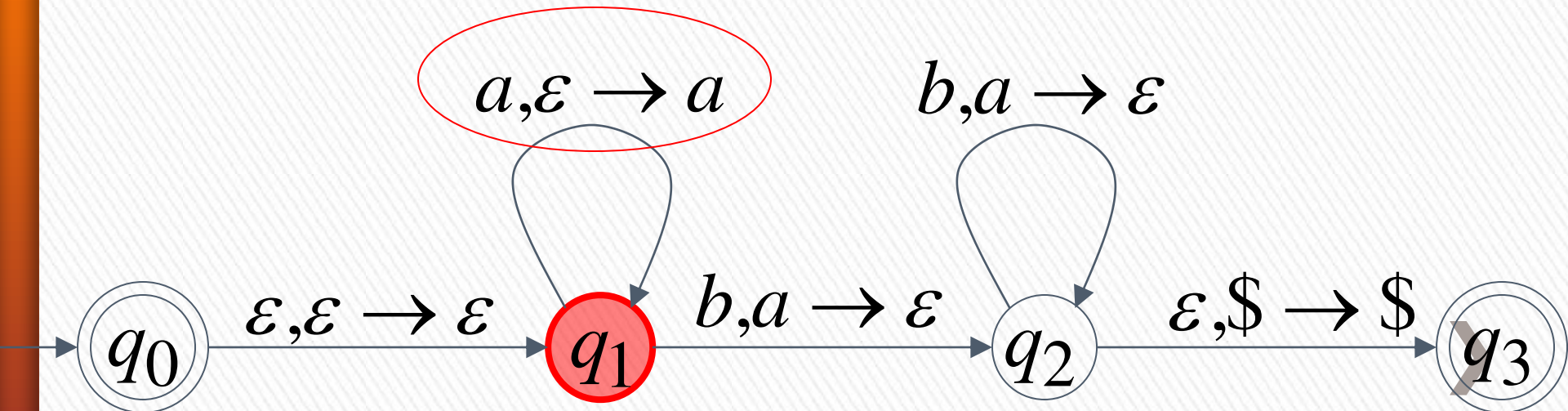


Time 2

Input

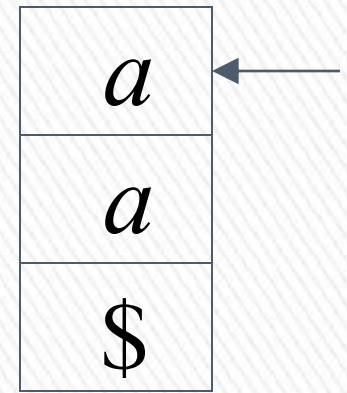
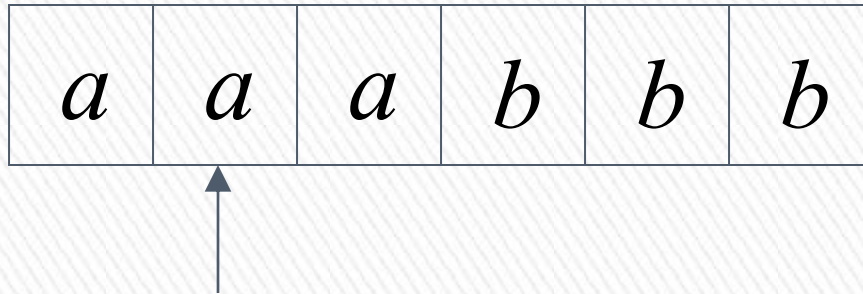


Stack

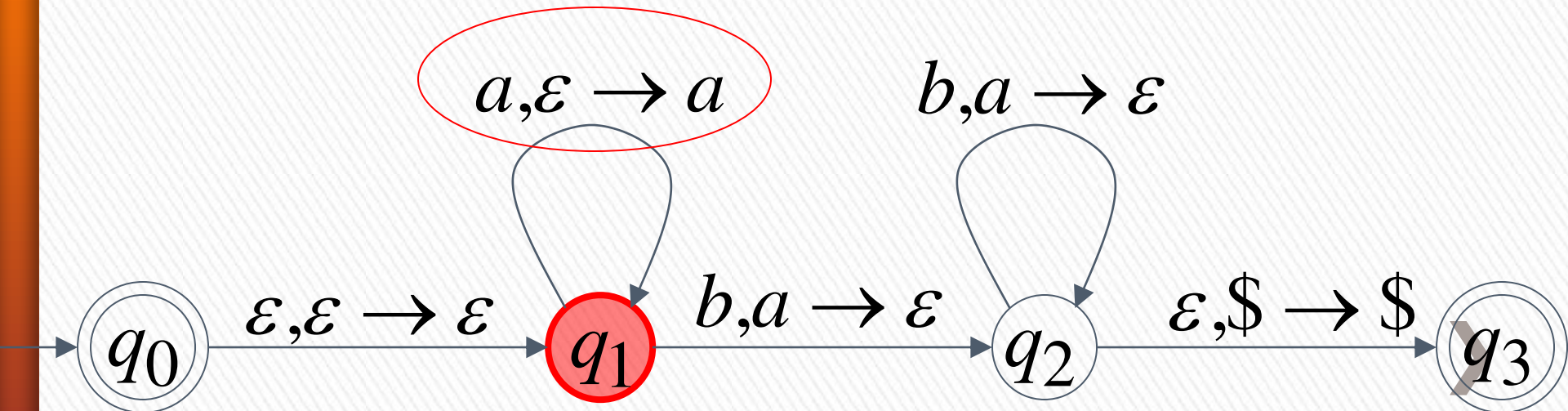


Time 3

Input

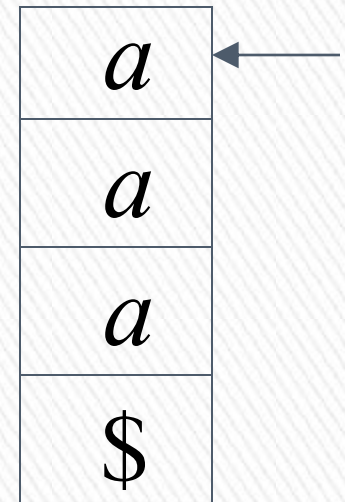


Stack

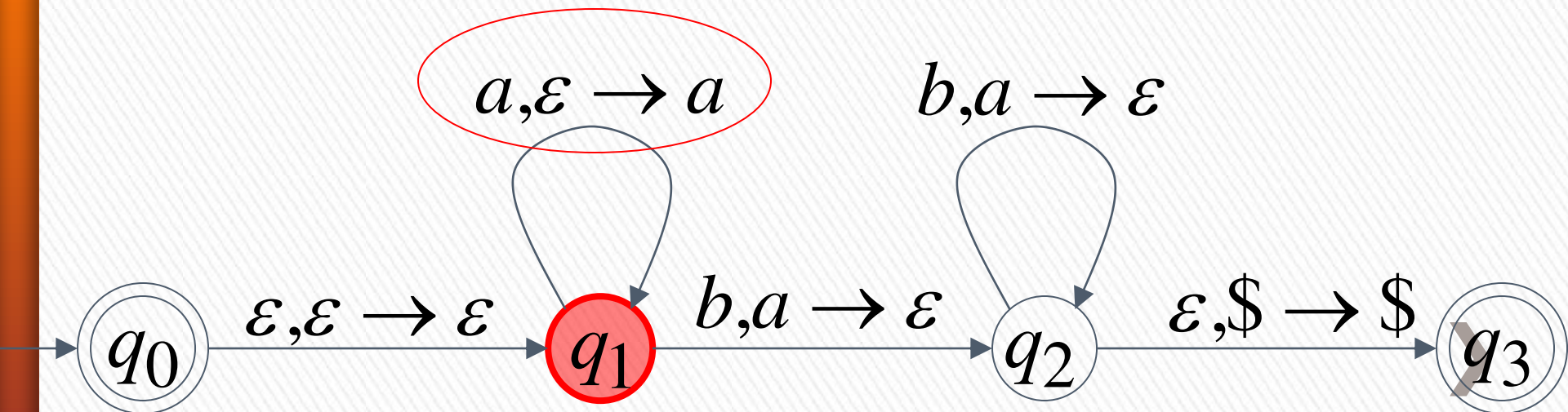


Time 4

Input

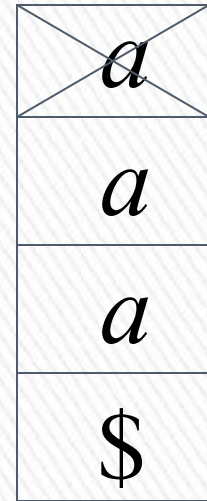


Stack

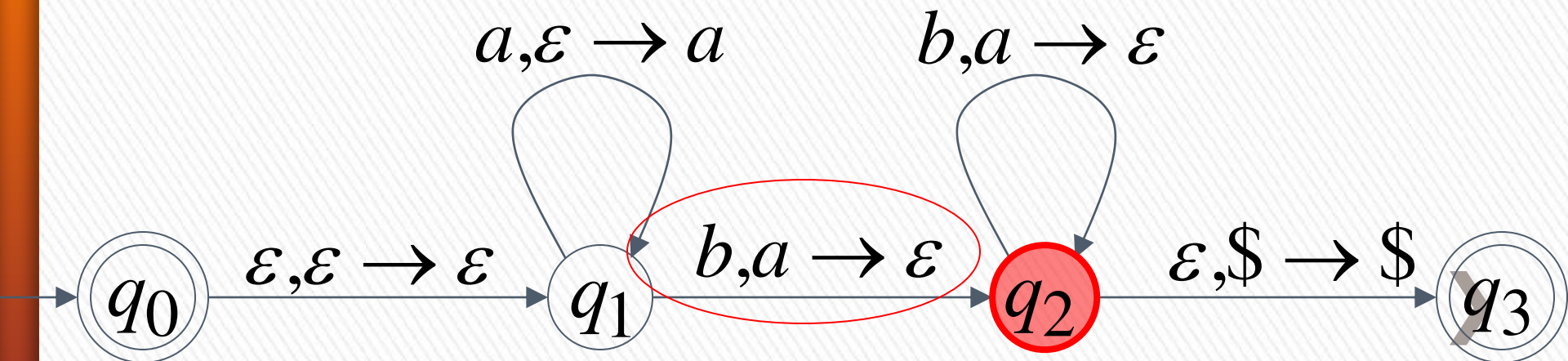


Time 5

Input

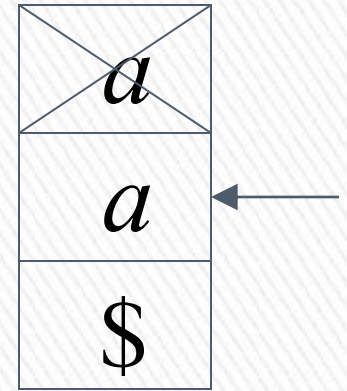


Stack

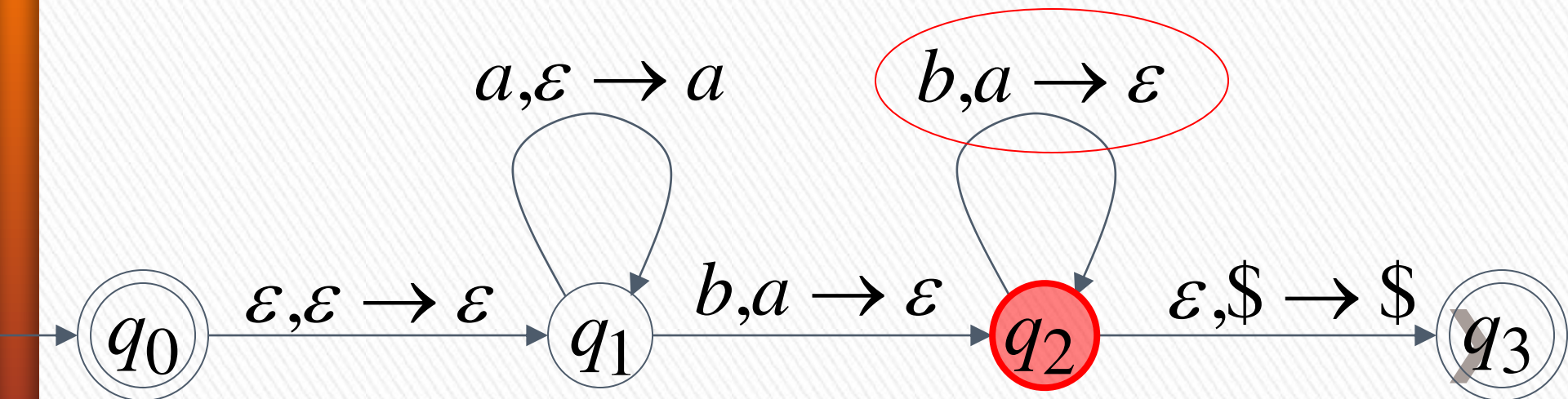


Time 6

Input

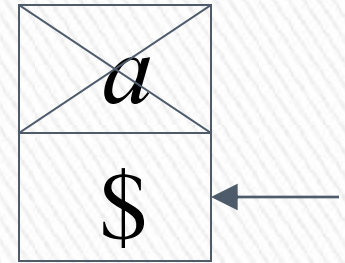
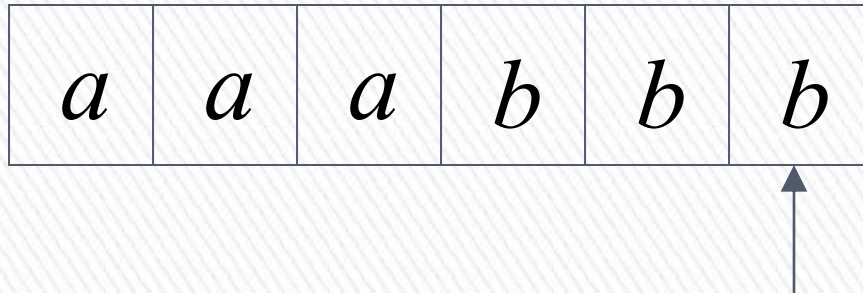


Stack

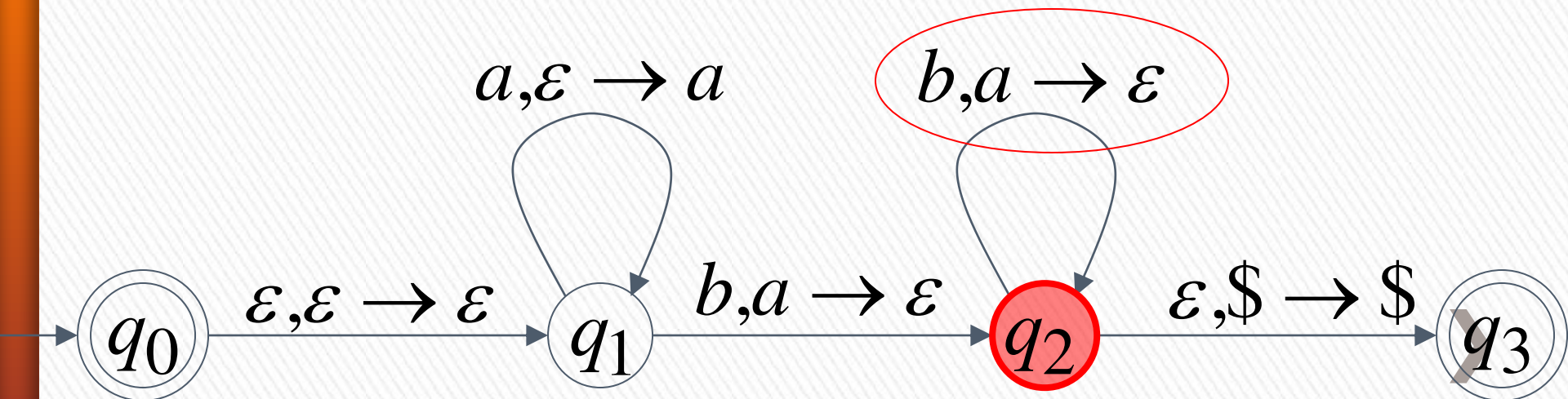


Time 7

Input

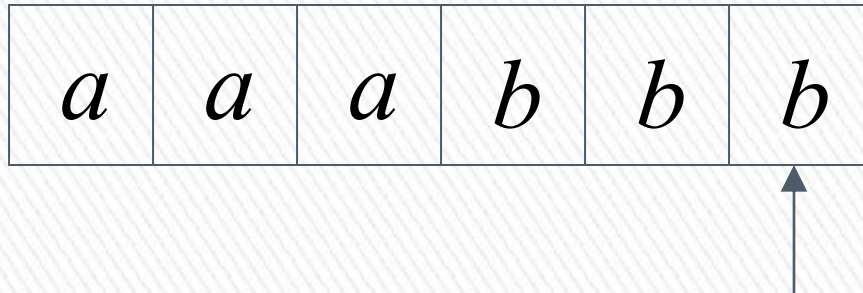


Stack

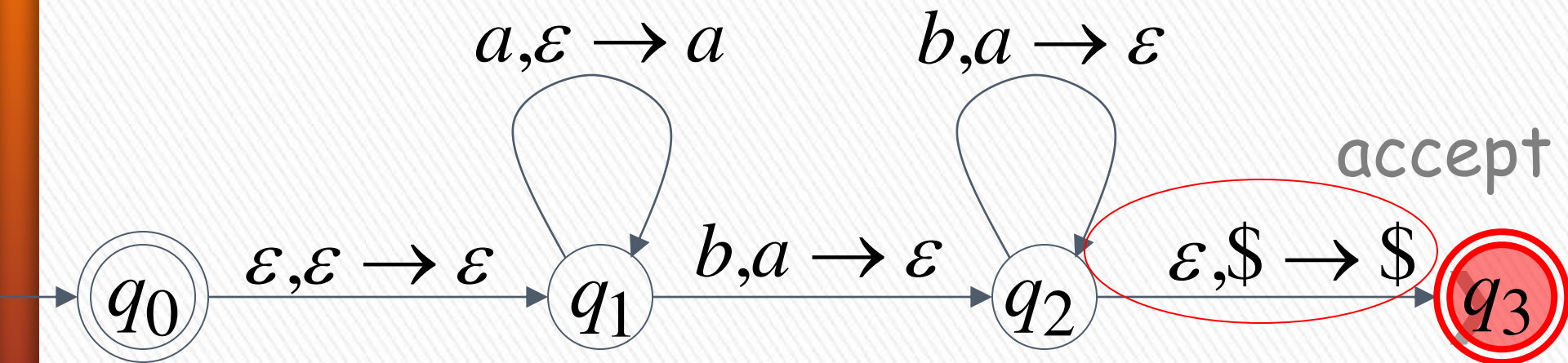


Time 8

Input



Stack



A string is accepted if there is
a computation such that:

All the input is consumed

AND

The last state is an accepting state

we do not care about the stack contents
at the end of the accepting computation ➤

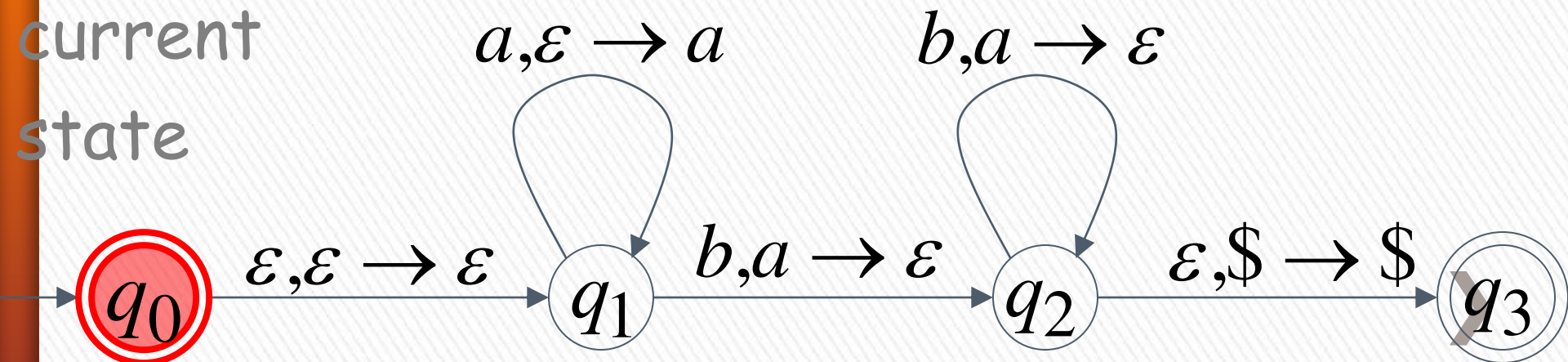
Rejection Example: Time 0

Input



Stack

current
state



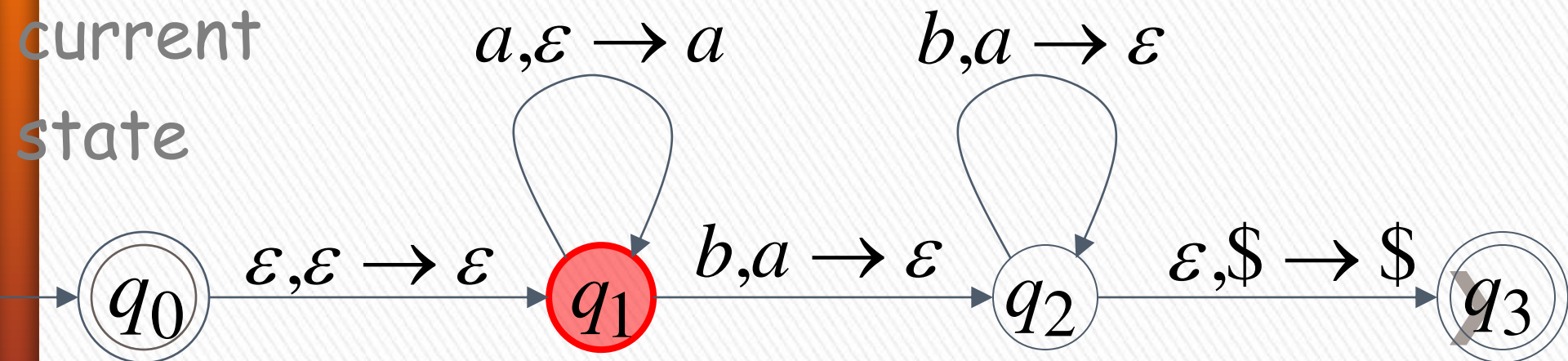
Rejection Example: Time 1

Input



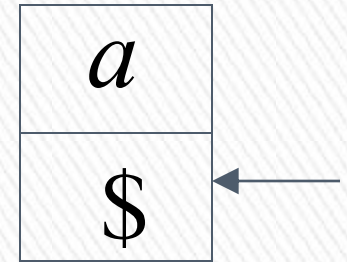
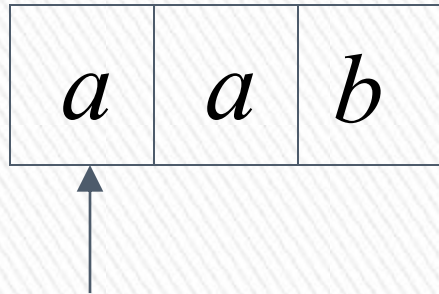
Stack

current
state



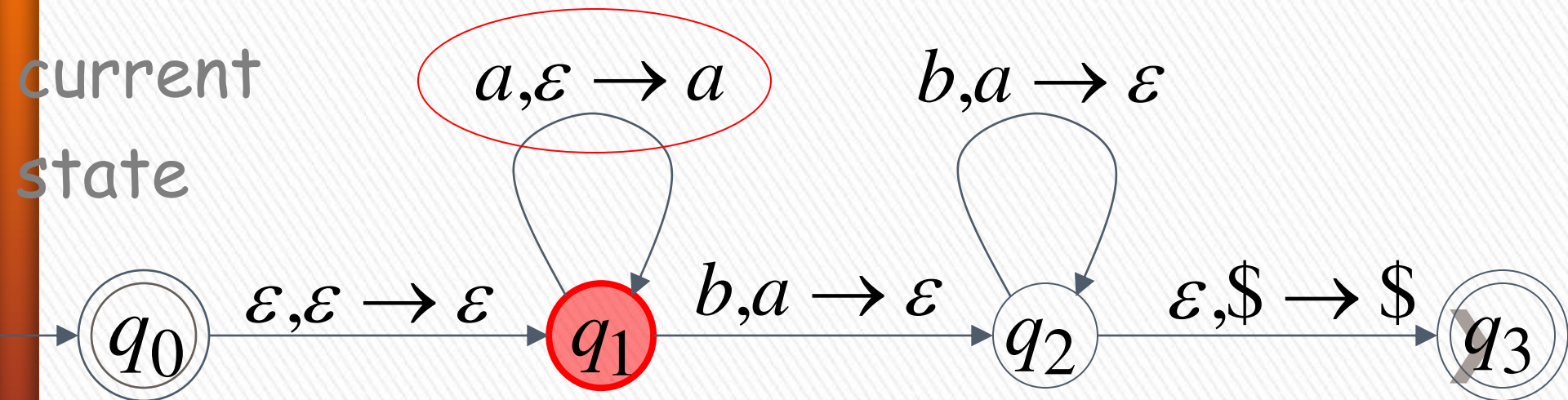
Rejection Example: Time 2

Input



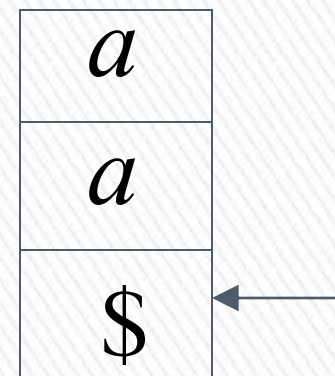
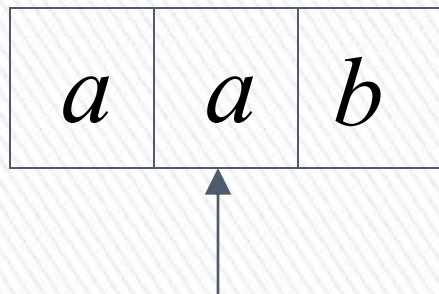
Stack

current
state



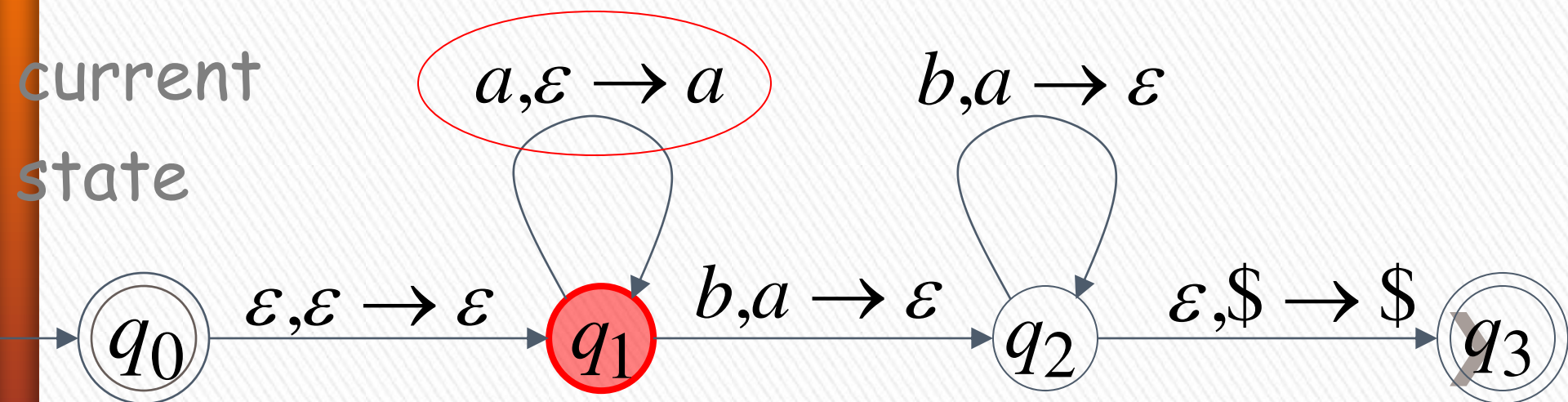
Rejection Example: Time 3

Input



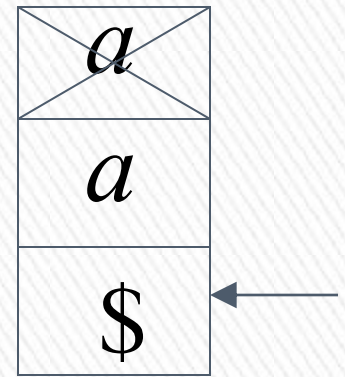
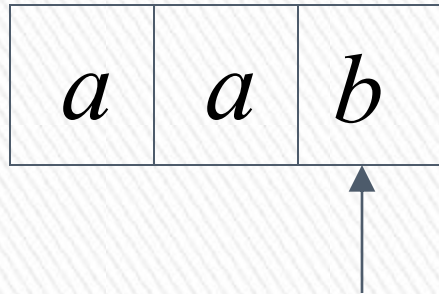
Stack

current
state



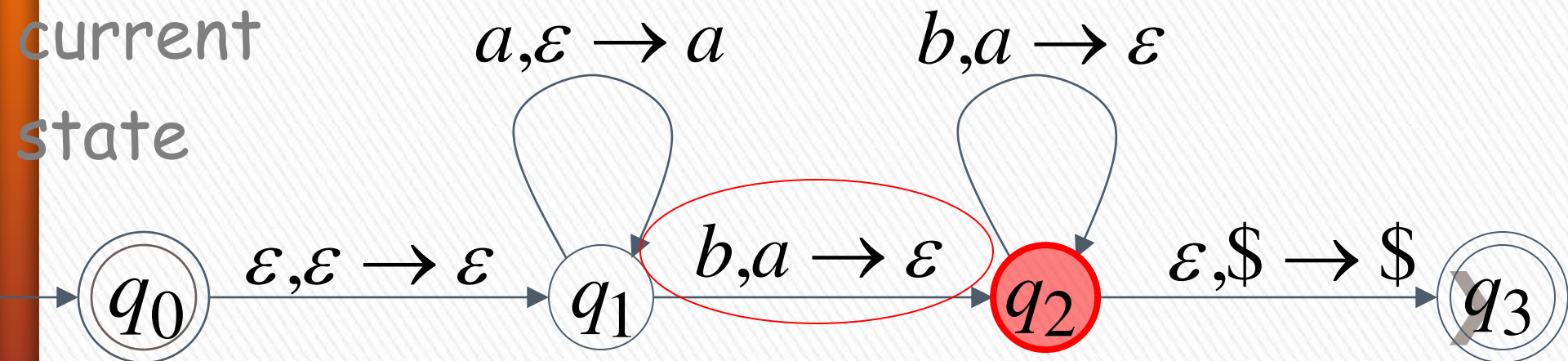
Rejection Example: Time 4

Input



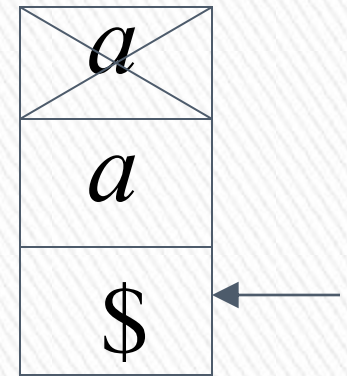
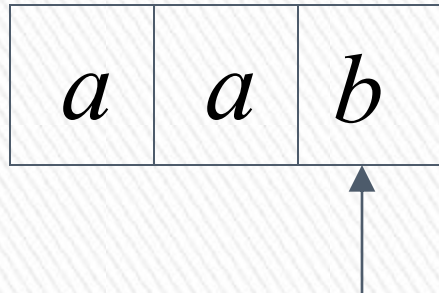
Stack

current
state



Rejection Example: Time 4

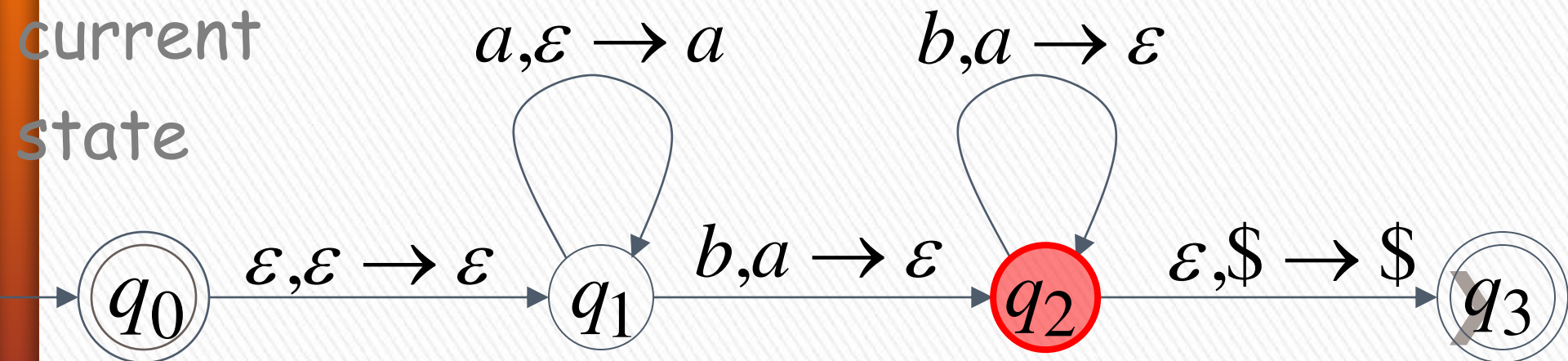
Input



Stack

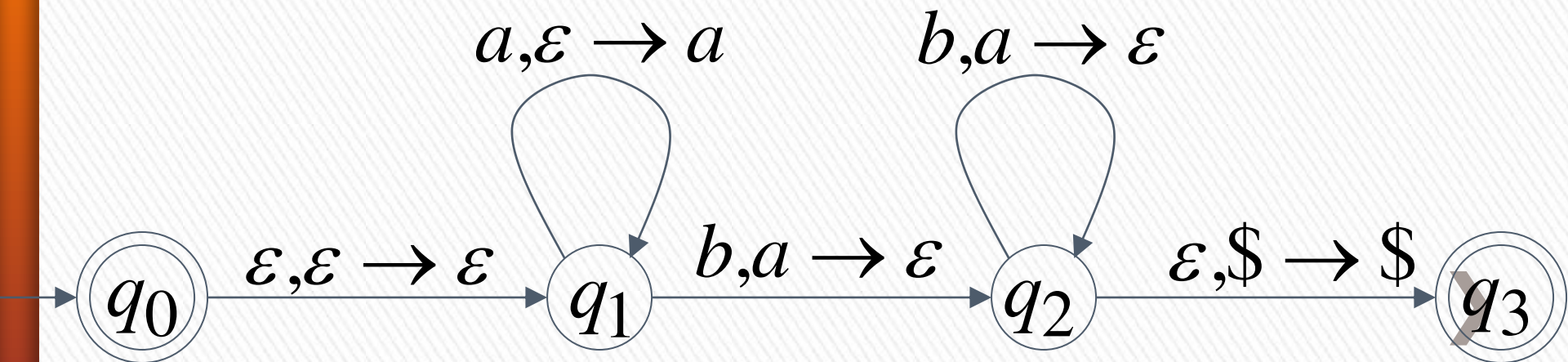
reject

current
state



There is no accepting computation for aab

The string aab is rejected by the PDA



Another PDA example

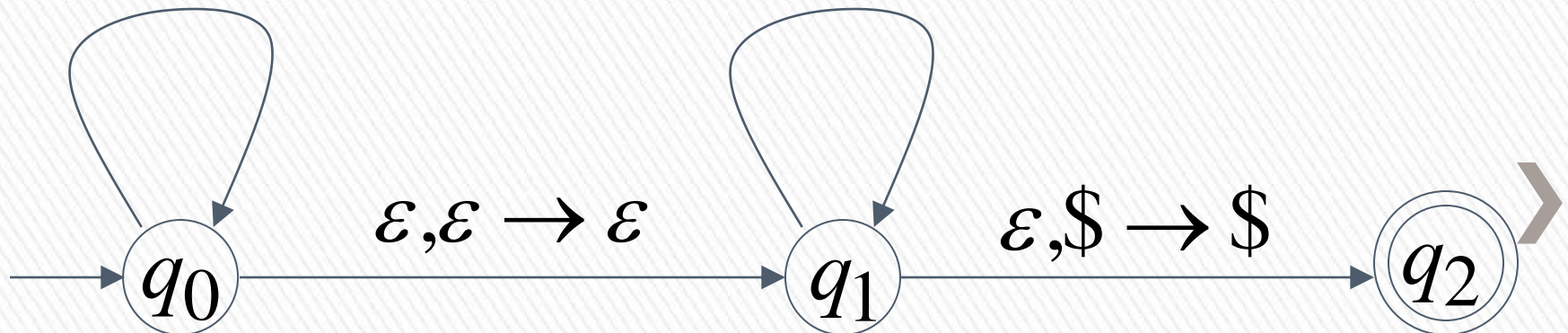
PDA M : $L(M) = \{vv^R : v \in \{a,b\}^*\}$

$a, \varepsilon \rightarrow a$

$a, a \rightarrow \varepsilon$

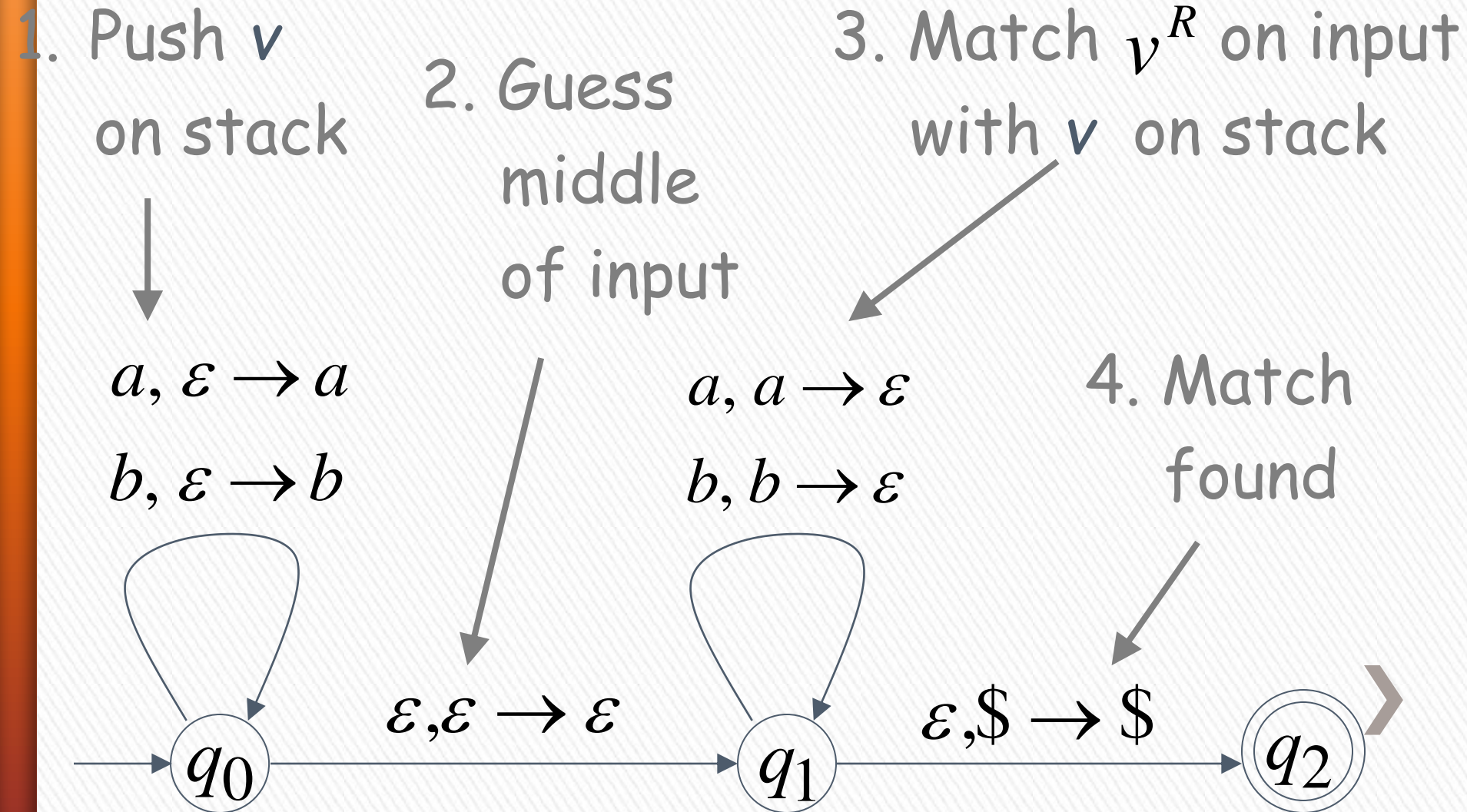
$b, \varepsilon \rightarrow b$

$b, b \rightarrow \varepsilon$



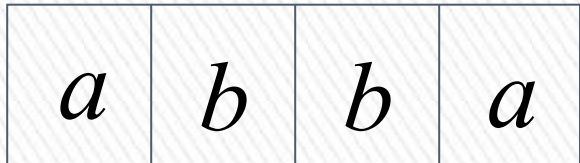
Basic Idea:

$$L(M) = \{vv^R : v \in \{a,b\}^*\}$$



Execution Example: Time 0

Input



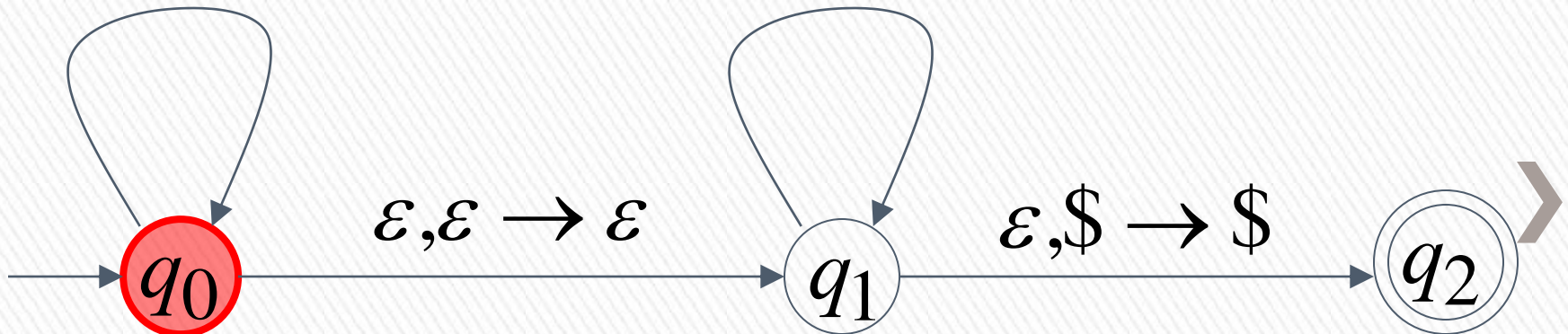
Stack

$a, \varepsilon \rightarrow a$

$a, a \rightarrow \varepsilon$

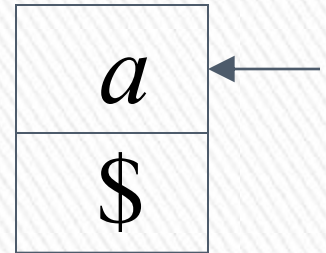
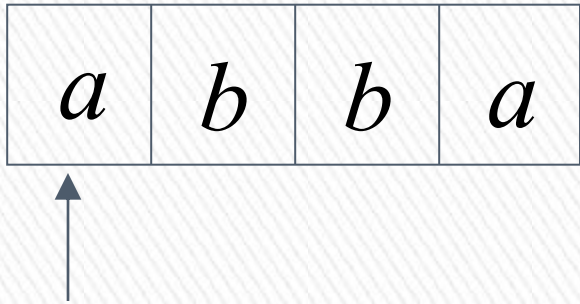
$b, \varepsilon \rightarrow b$

$b, b \rightarrow \varepsilon$



Time 1

Input



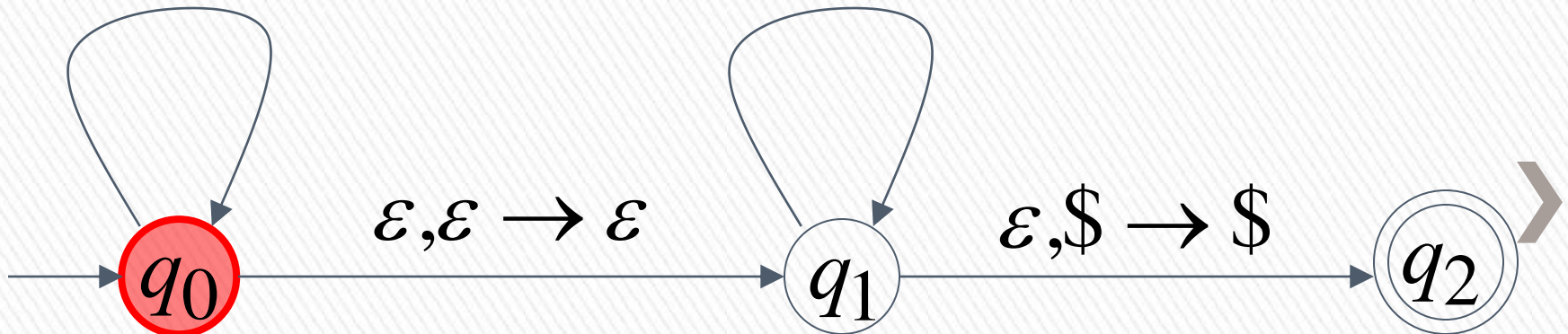
Stack

$a, \varepsilon \rightarrow a$

$b, \varepsilon \rightarrow b$

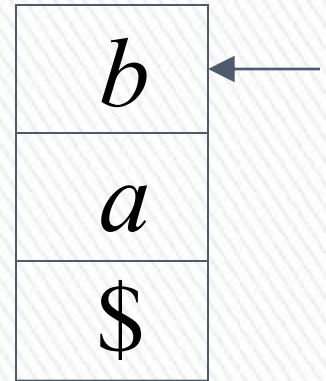
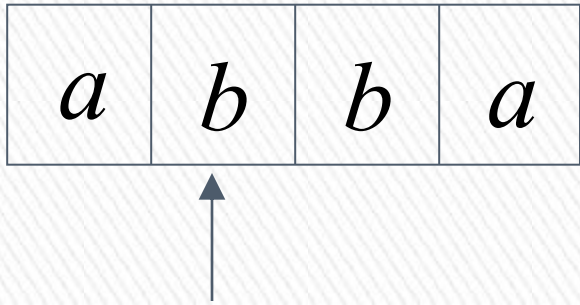
$a, a \rightarrow \varepsilon$

$b, b \rightarrow \varepsilon$



Time 2

Input



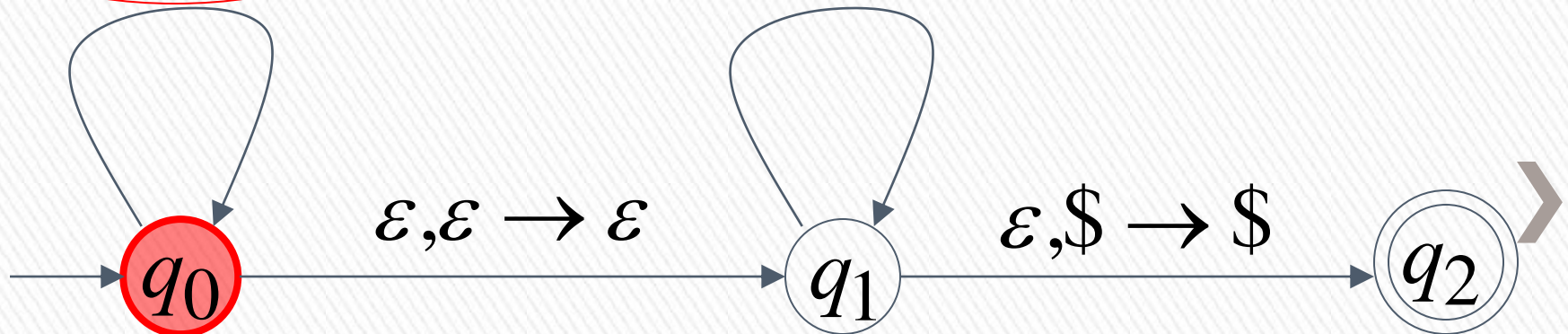
Stack

$a, \varepsilon \rightarrow a$

$b, \varepsilon \rightarrow b$

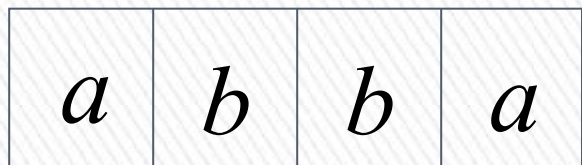
$a, a \rightarrow \varepsilon$

$b, b \rightarrow \varepsilon$

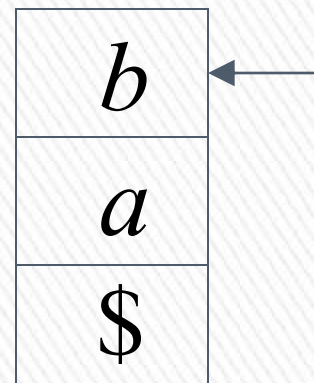


Time 3

Input



Guess the middle
of string



Stack

$a, \varepsilon \rightarrow a$

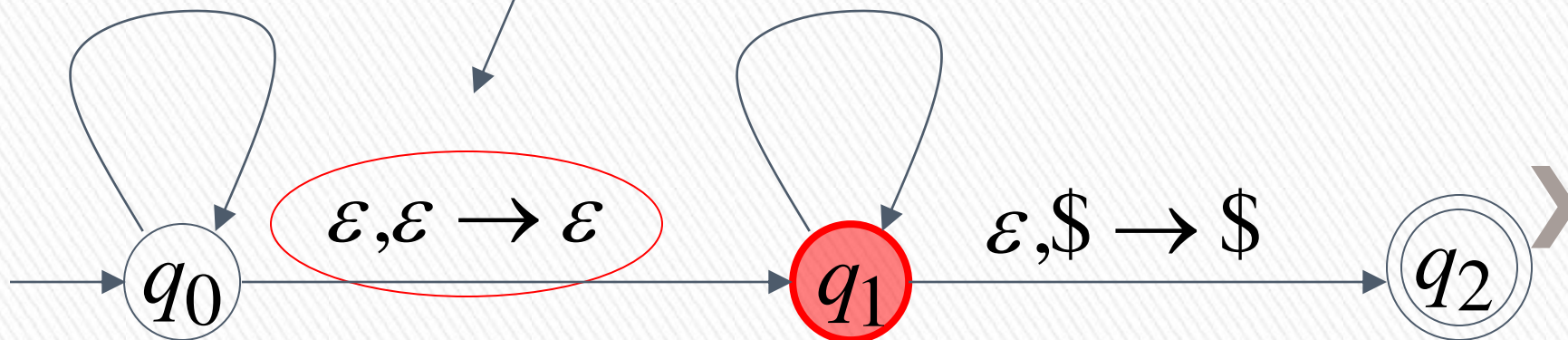
$b, \varepsilon \rightarrow b$

$a, a \rightarrow \varepsilon$

$b, b \rightarrow \varepsilon$

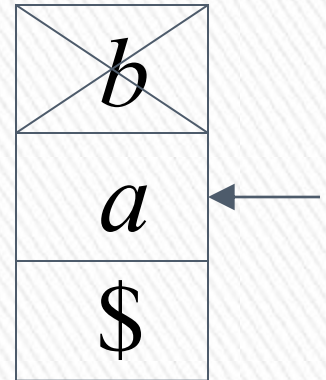
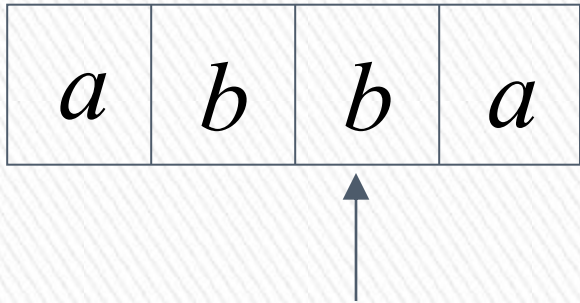
$\varepsilon, \varepsilon \rightarrow \varepsilon$

$\varepsilon, \$ \rightarrow \$$



Time 4

Input



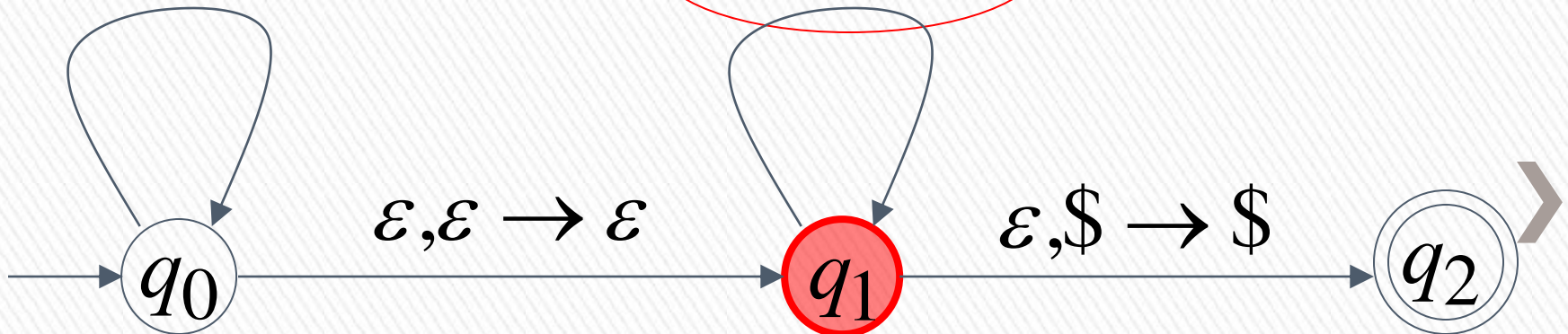
Stack

$a, \varepsilon \rightarrow a$

$b, \varepsilon \rightarrow b$

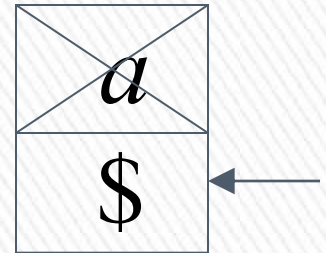
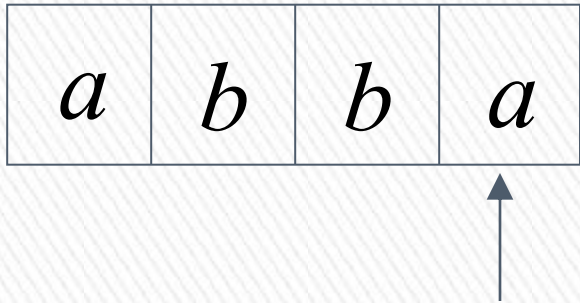
$a, a \rightarrow \varepsilon$

$b, b \rightarrow \varepsilon$



Time 5

Input



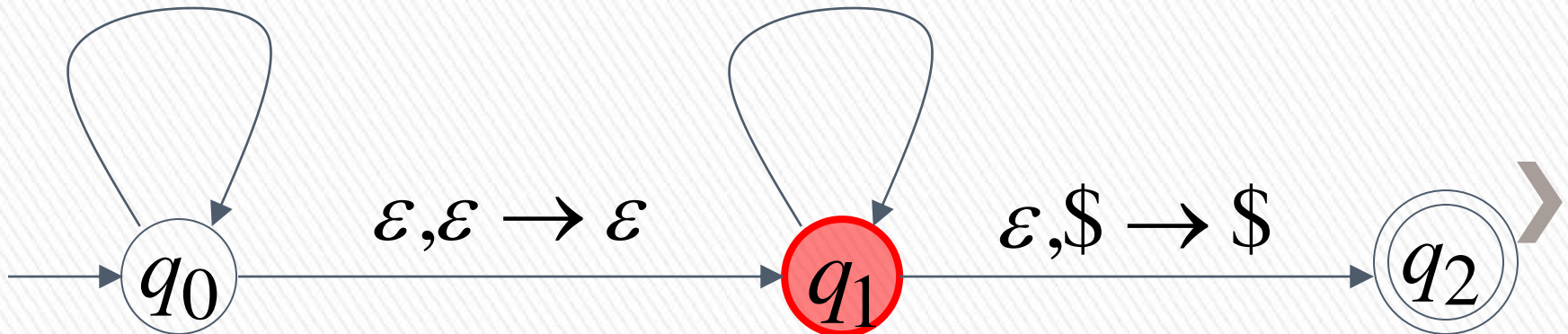
Stack

$a, \varepsilon \rightarrow a$

$b, \varepsilon \rightarrow b$

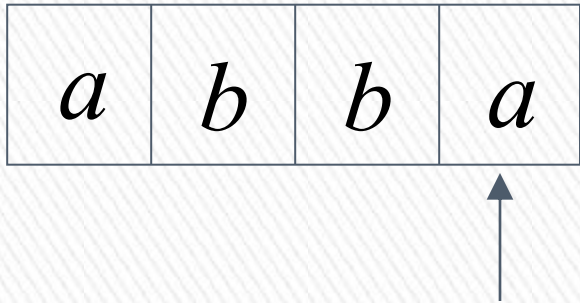
$a, a \rightarrow \varepsilon$

$b, b \rightarrow \varepsilon$



Time 6

Input



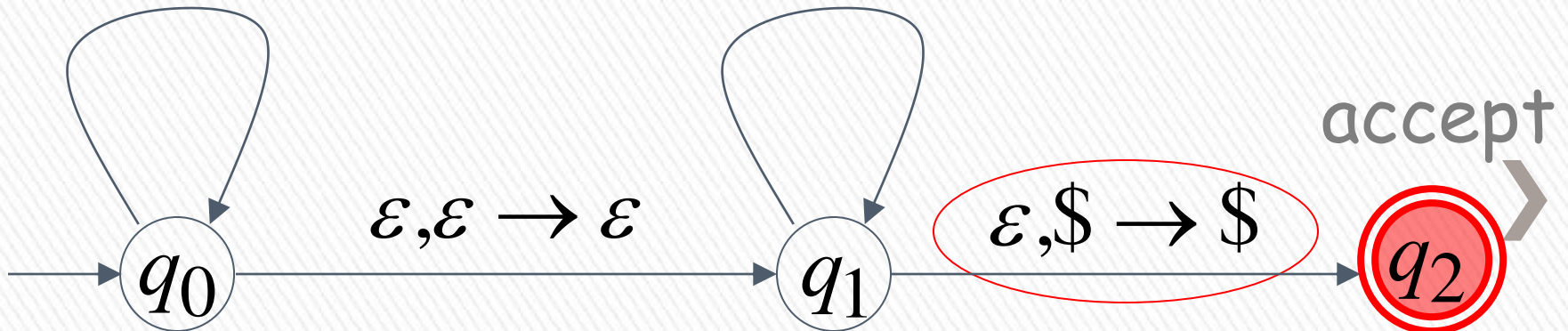
Stack

$a, \varepsilon \rightarrow a$

$a, a \rightarrow \varepsilon$

$b, \varepsilon \rightarrow b$

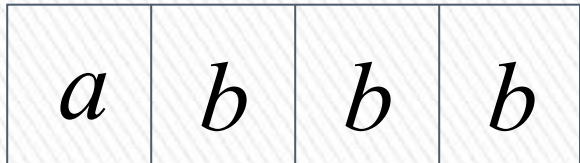
$b, b \rightarrow \varepsilon$



Rejection Example:

Time 0

Input



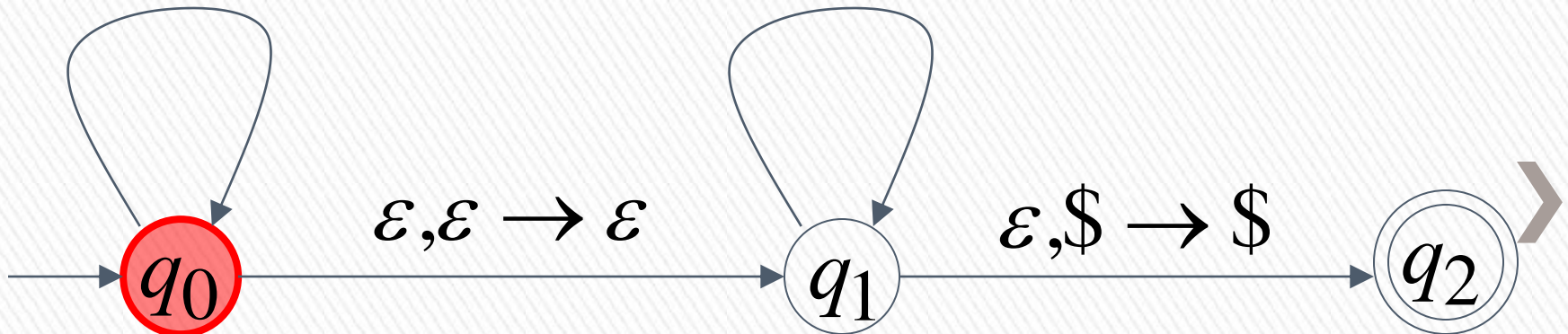
Stack

$a, \varepsilon \rightarrow a$

$a, a \rightarrow \varepsilon$

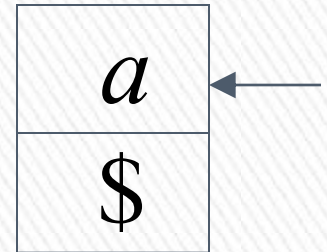
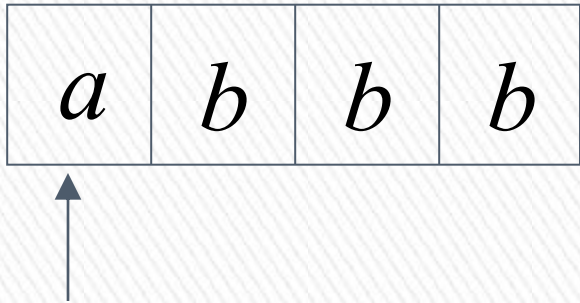
$b, \varepsilon \rightarrow b$

$b, b \rightarrow \varepsilon$



Time 1

Input



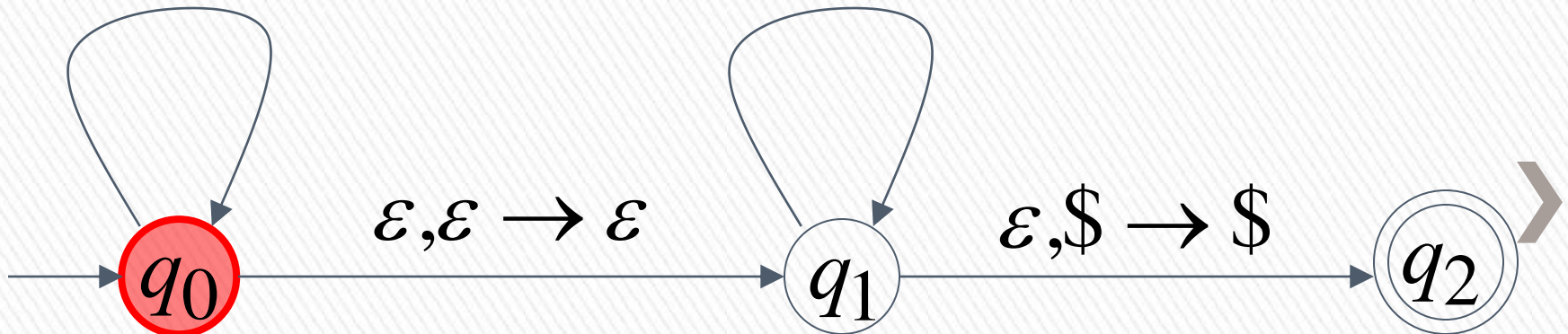
Stack

$a, \varepsilon \rightarrow a$

$b, \varepsilon \rightarrow b$

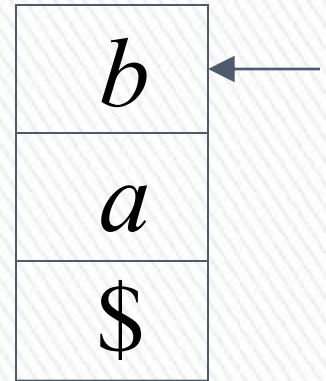
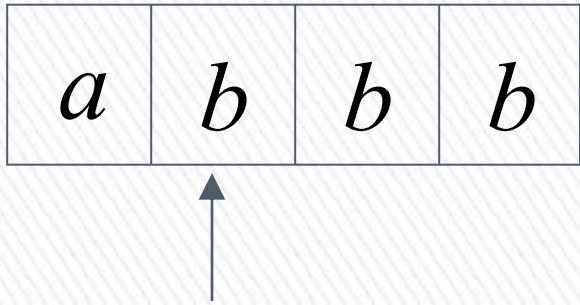
$a, a \rightarrow \varepsilon$

$b, b \rightarrow \varepsilon$



Time 2

Input



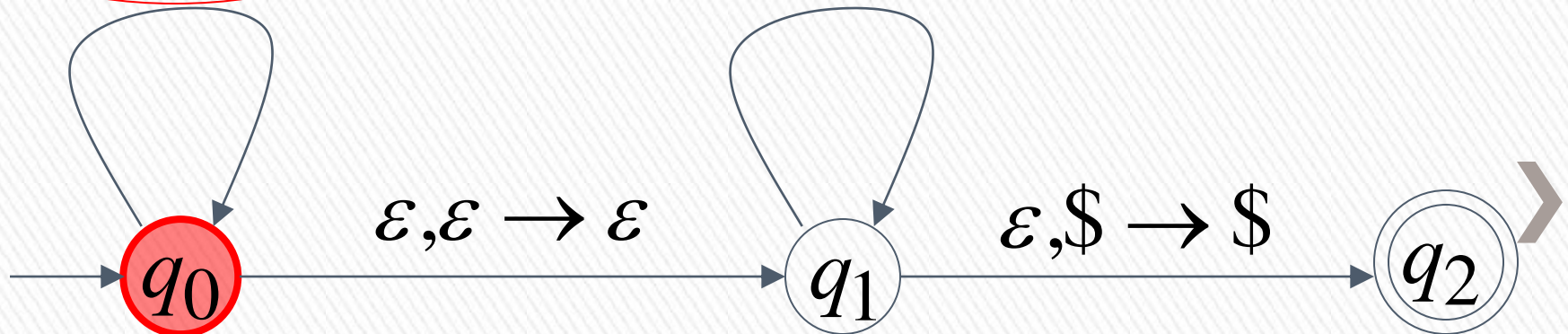
Stack

$a, \varepsilon \rightarrow a$

$b, \varepsilon \rightarrow b$

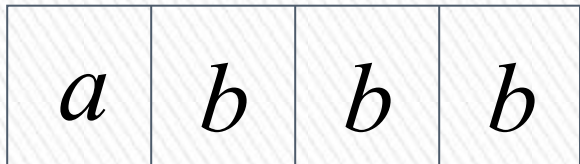
$a, a \rightarrow \varepsilon$

$b, b \rightarrow \varepsilon$

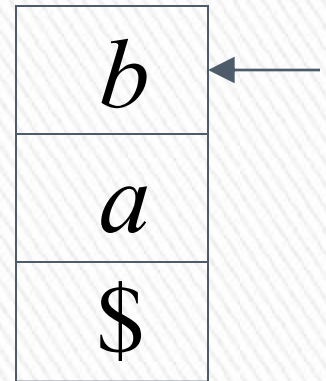


Time 3

Input



Guess the middle
of string



Stack

$a, \varepsilon \rightarrow a$

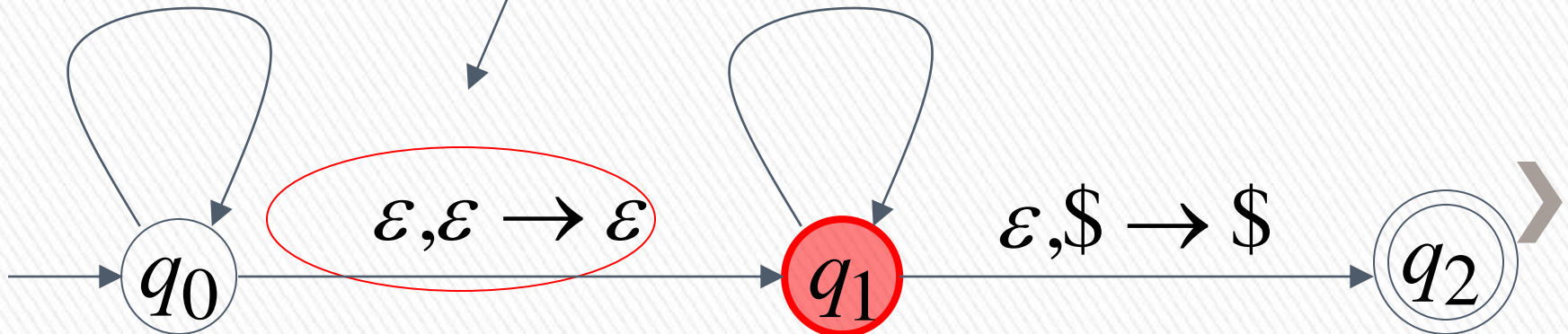
$b, \varepsilon \rightarrow b$

$a, a \rightarrow \varepsilon$

$b, b \rightarrow \varepsilon$

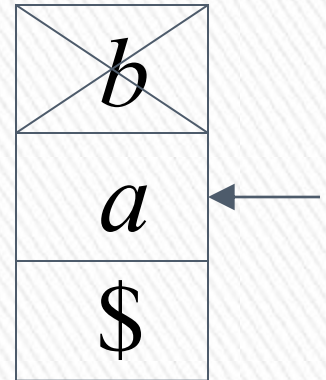
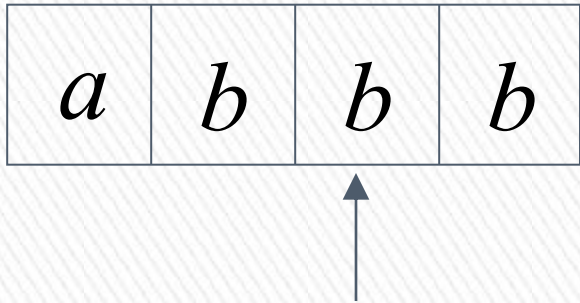
$\varepsilon, \varepsilon \rightarrow \varepsilon$

$\varepsilon, \$ \rightarrow \$$



Time 4

Input



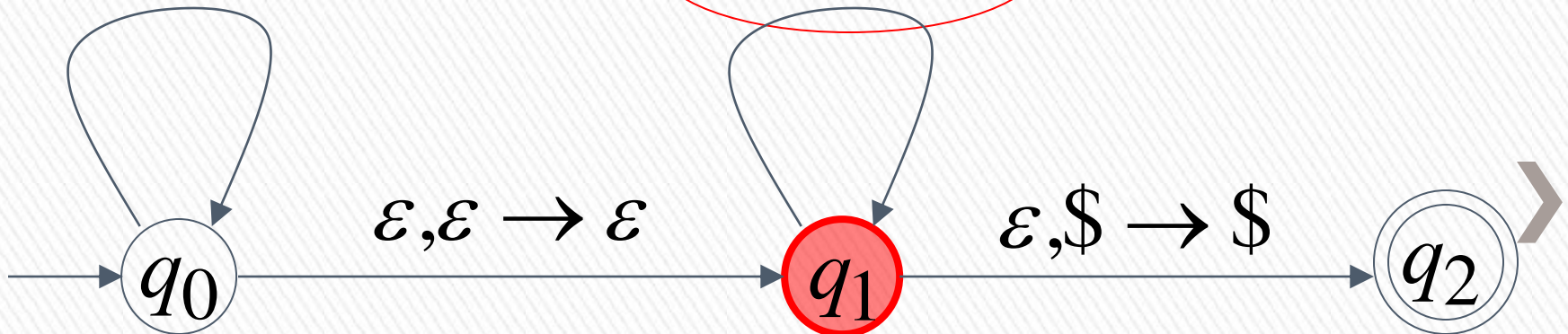
Stack

$a, \varepsilon \rightarrow a$

$b, \varepsilon \rightarrow b$

$a, a \rightarrow \varepsilon$

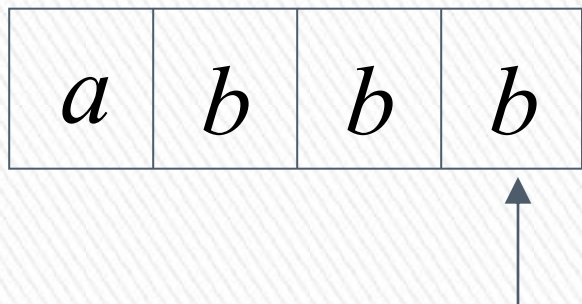
$b, b \rightarrow \varepsilon$



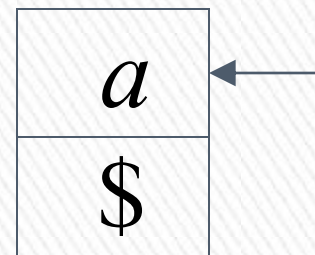
Time 5

Input

There is no possible transition.



Input is not consumed



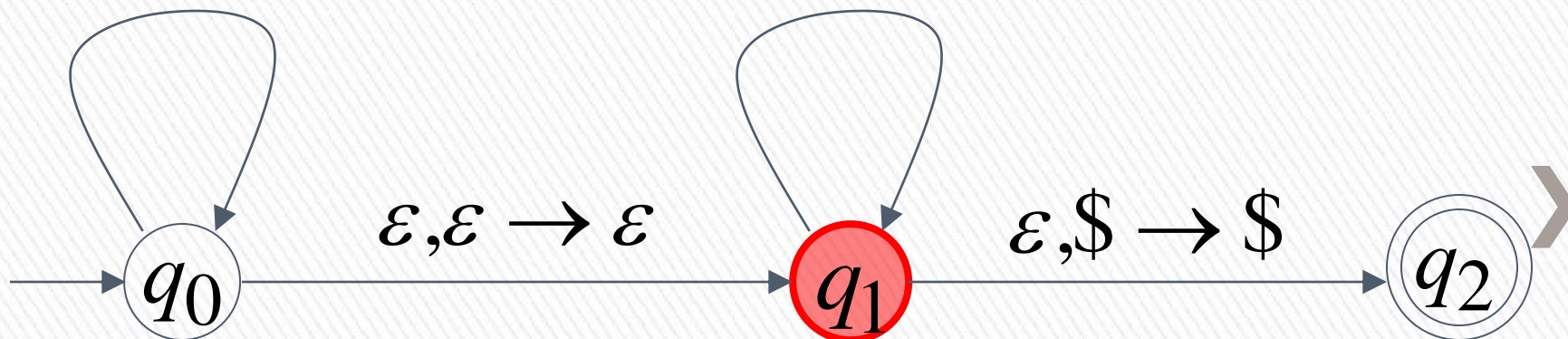
Stack

$a, \varepsilon \rightarrow a$

$a, a \rightarrow \varepsilon$

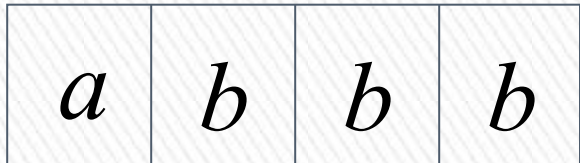
$b, \varepsilon \rightarrow b$

$b, b \rightarrow \varepsilon$



Another computation on same string:

Input



Time 0



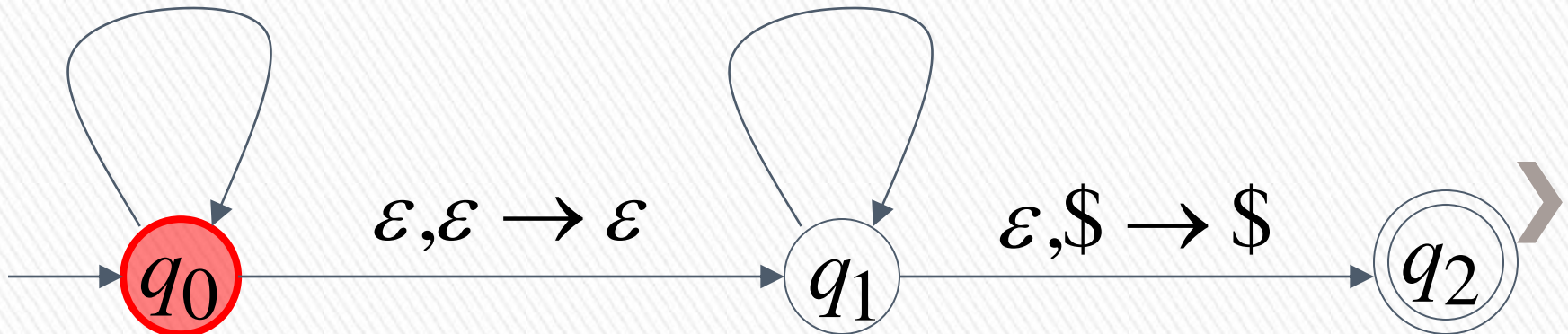
Stack

$a, \varepsilon \rightarrow a$

$a, a \rightarrow \varepsilon$

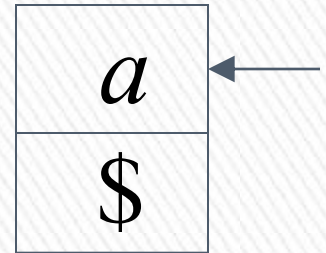
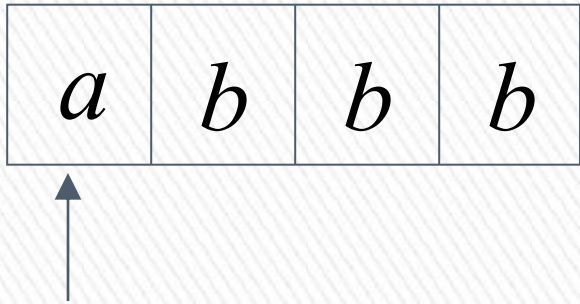
$b, \varepsilon \rightarrow b$

$b, b \rightarrow \varepsilon$



Time 1

Input



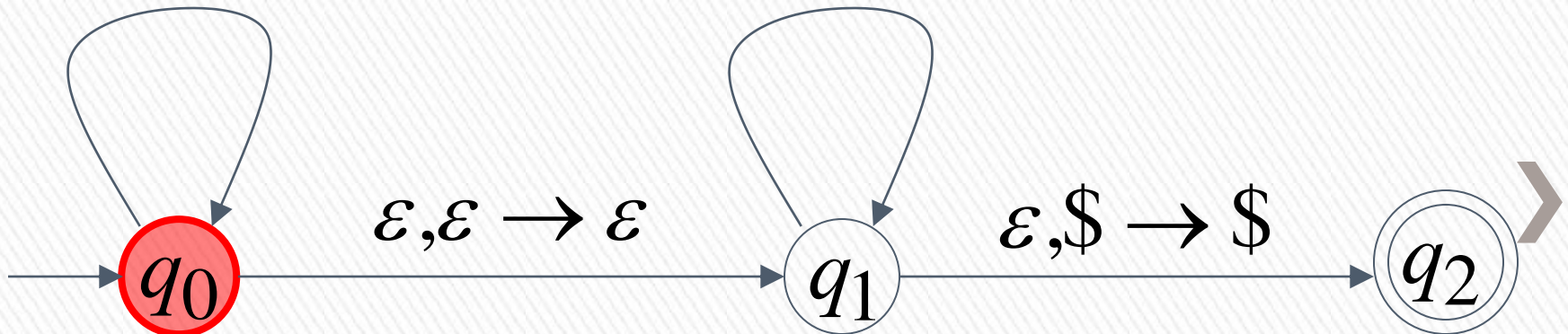
Stack

$a, \varepsilon \rightarrow a$

$b, \varepsilon \rightarrow b$

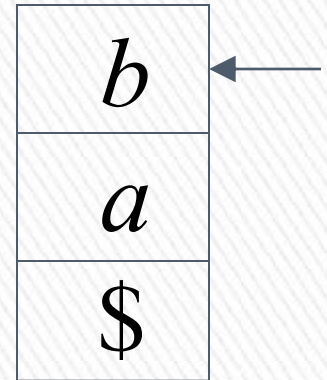
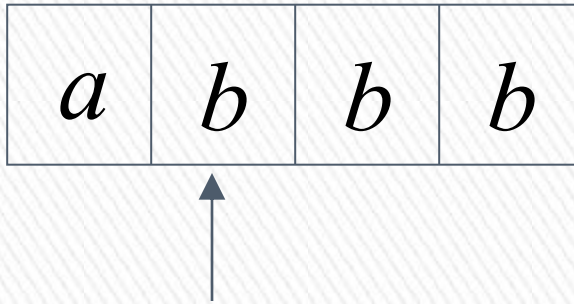
$a, a \rightarrow \varepsilon$

$b, b \rightarrow \varepsilon$



Time 2

Input



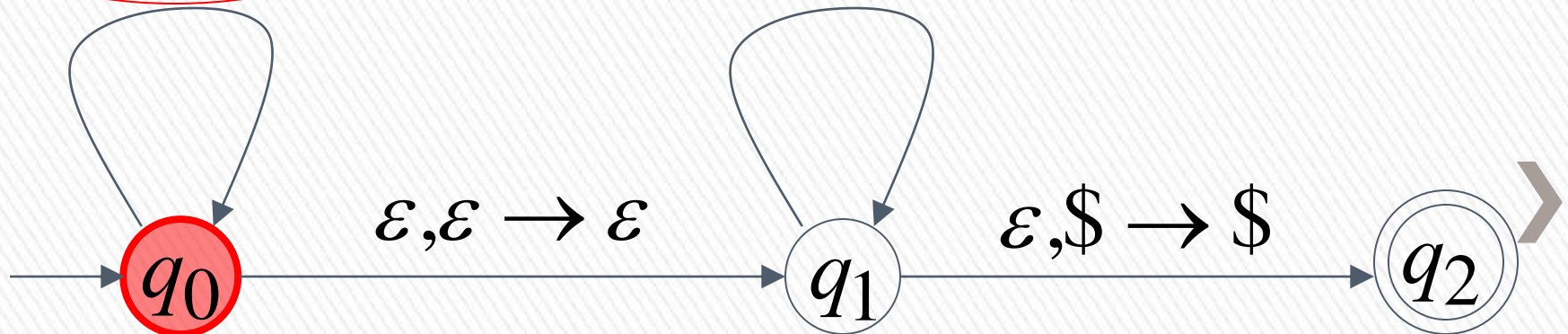
Stack

$a, \varepsilon \rightarrow a$

$b, \varepsilon \rightarrow b$

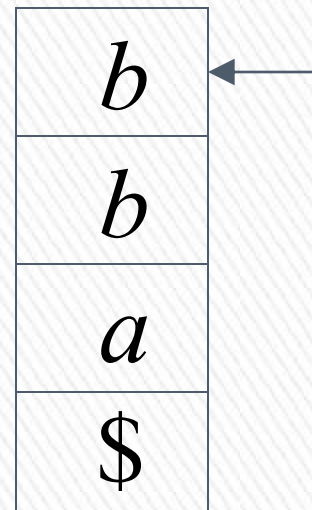
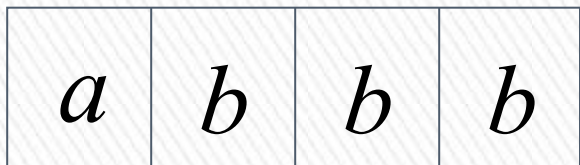
$a, a \rightarrow \varepsilon$

$b, b \rightarrow \varepsilon$



Time 3

Input



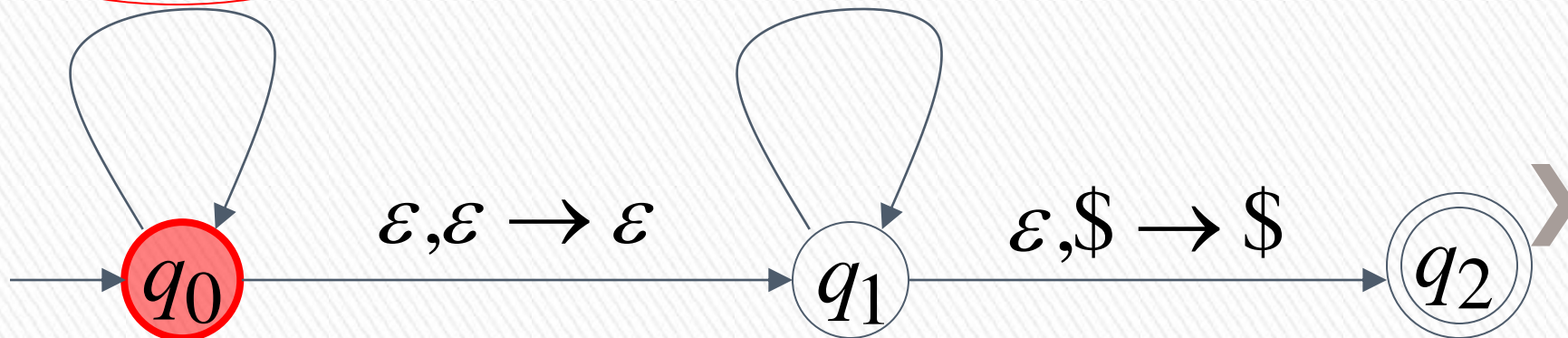
Stack

$a, \varepsilon \rightarrow a$

$b, \varepsilon \rightarrow b$

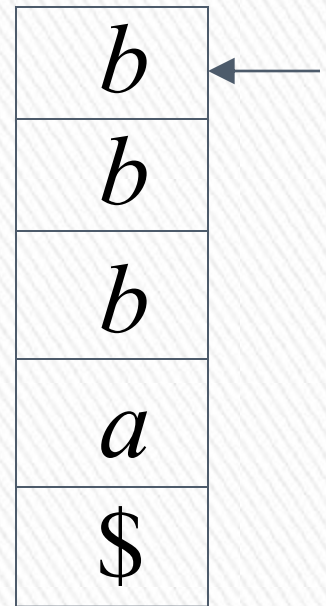
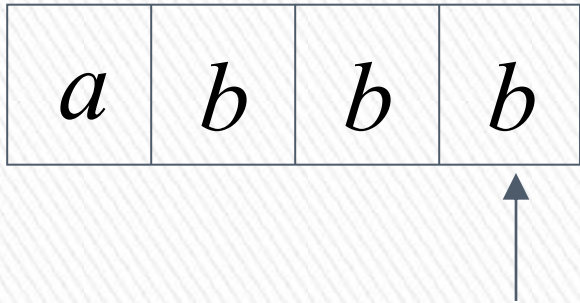
$a, a \rightarrow \varepsilon$

$b, b \rightarrow \varepsilon$



Time 4

Input



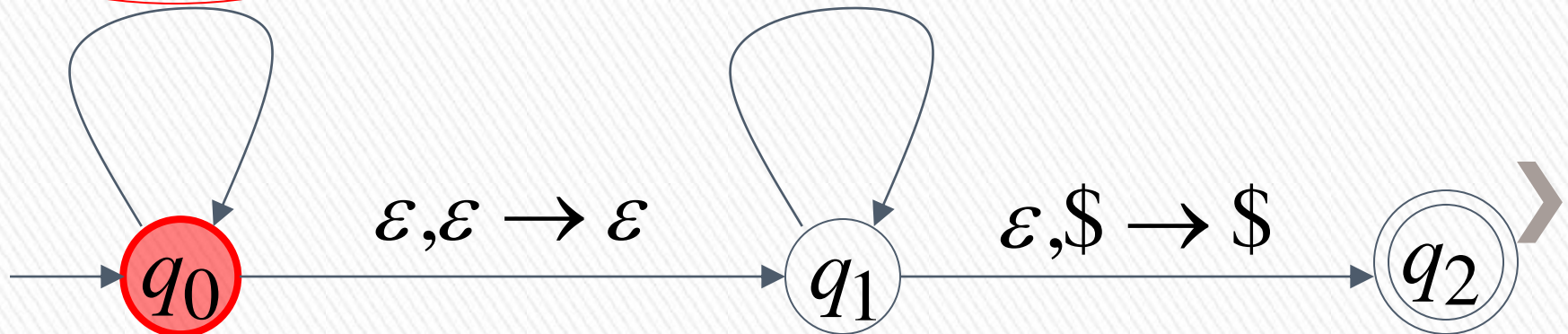
Stack

$a, \varepsilon \rightarrow a$

$b, \varepsilon \rightarrow b$

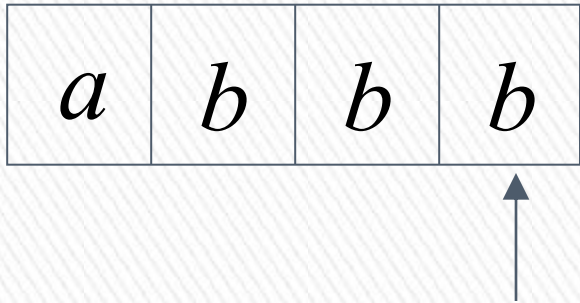
$a, a \rightarrow \varepsilon$

$b, b \rightarrow \varepsilon$

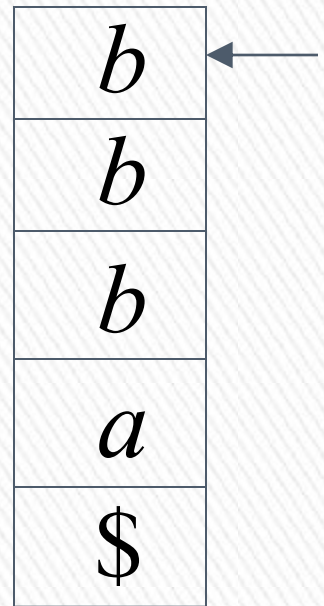


Time 5

Input



No accept state is reached



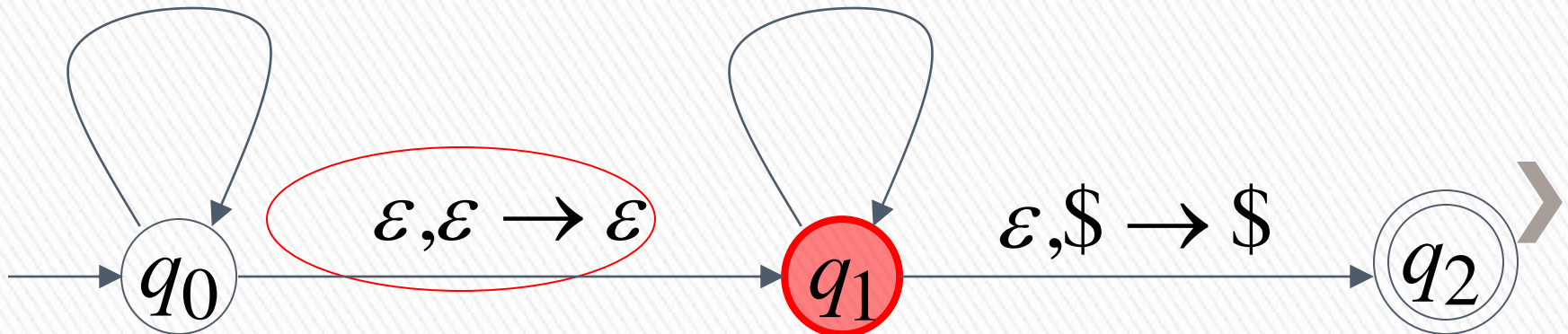
Stack

$a, \varepsilon \rightarrow a$

$b, \varepsilon \rightarrow b$

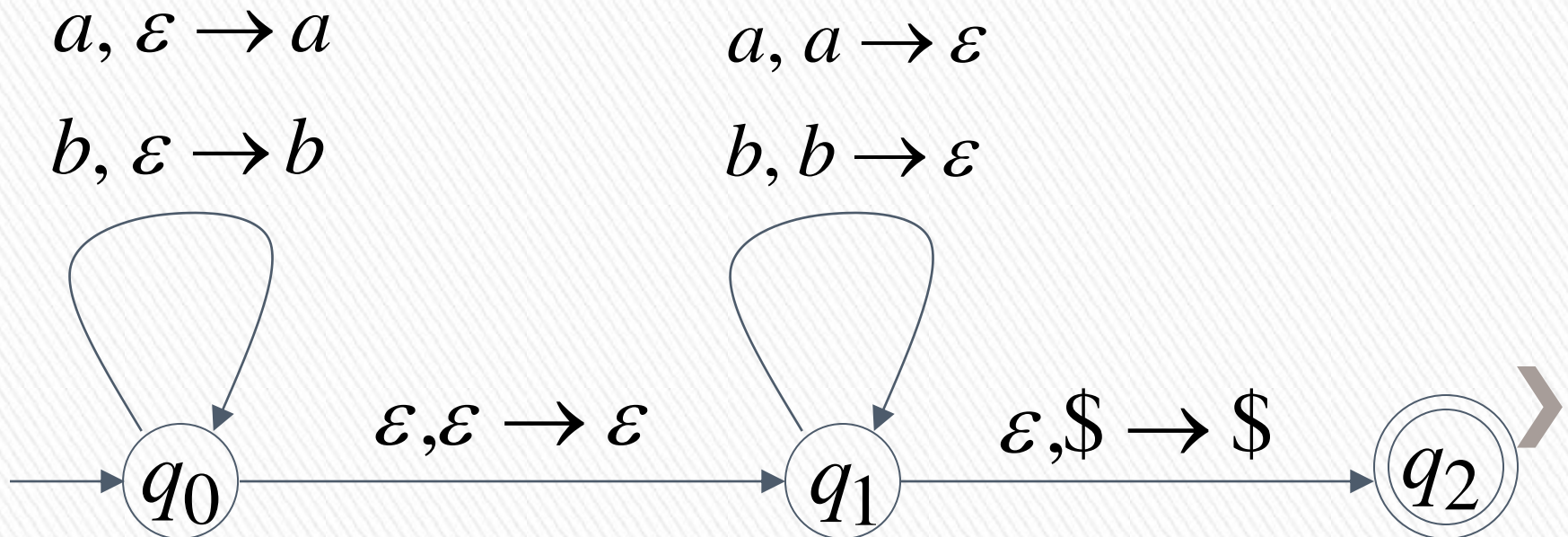
$a, a \rightarrow \varepsilon$

$b, b \rightarrow \varepsilon$

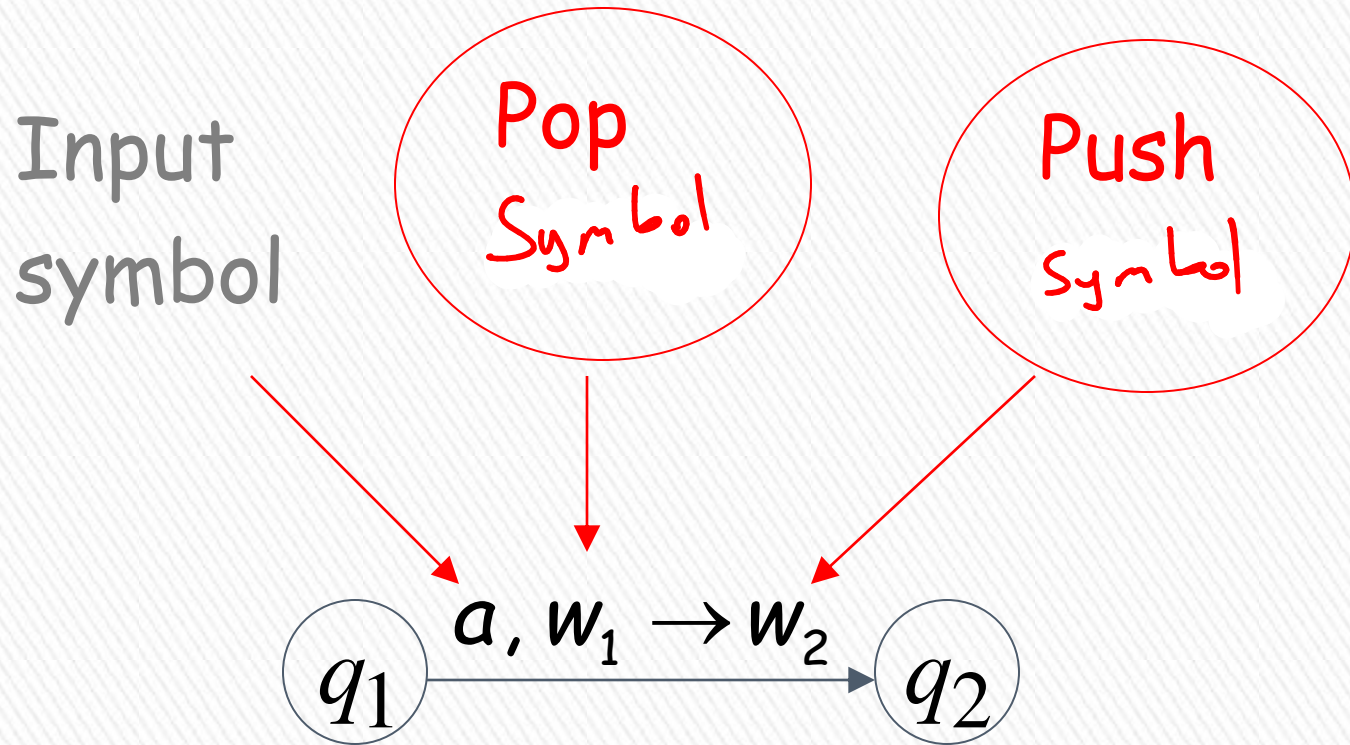


There is no computation
that accepts string $abbb$

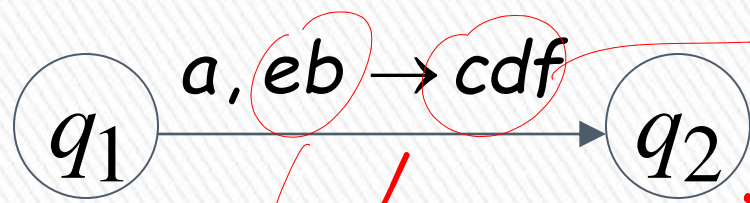
$$abbb \notin L(M)$$



Pushing & Popping Strings

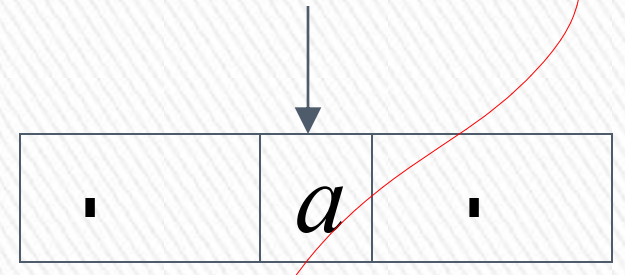
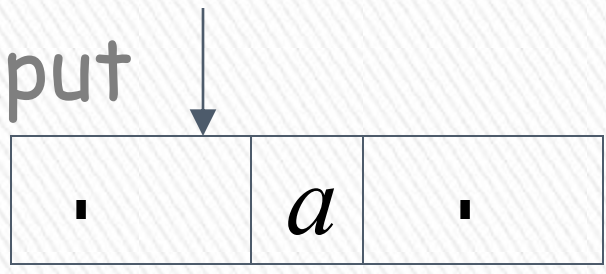


Example:

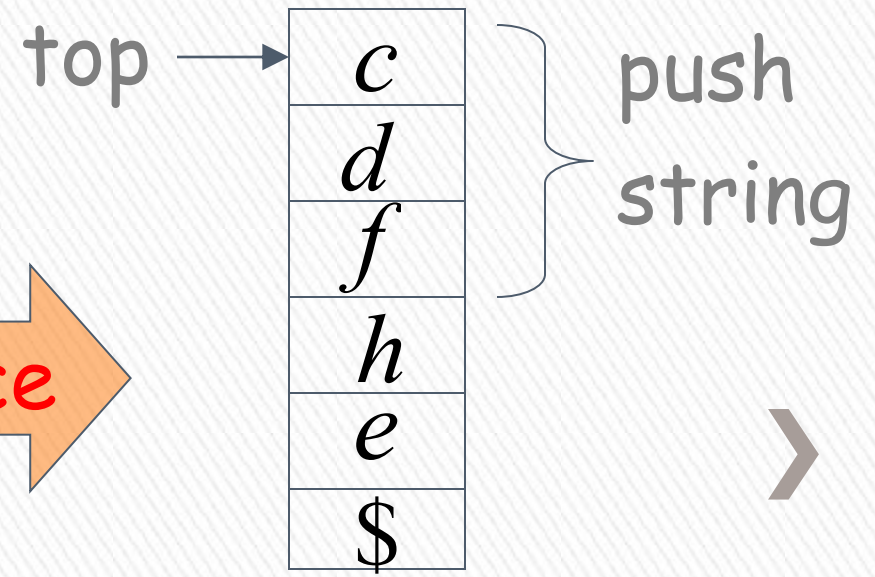
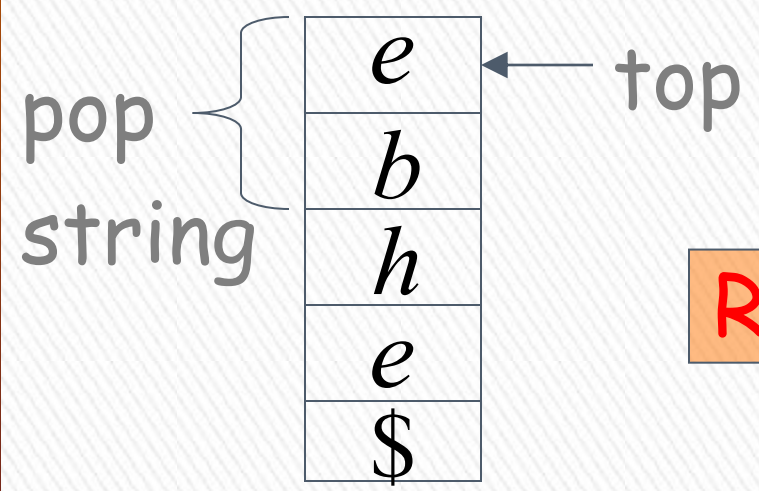


↳ *çizim kolaylığı*

input

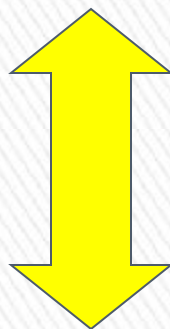
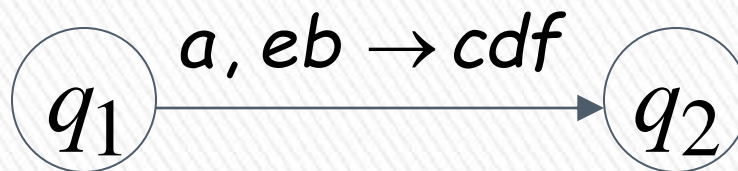


stack



Replace





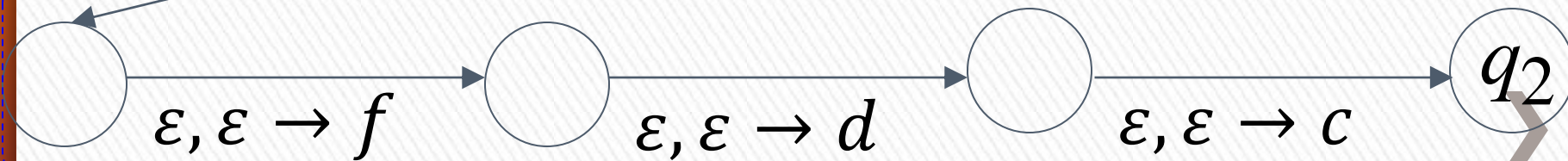
Equivalent
transitions

pop



$\varepsilon, \varepsilon \rightarrow \varepsilon$

push



Another PDA example

$$L(M) = \{w \in \{a,b\}^* : n_a(w) = n_b(w)\}$$

a says 's' - b says 's'

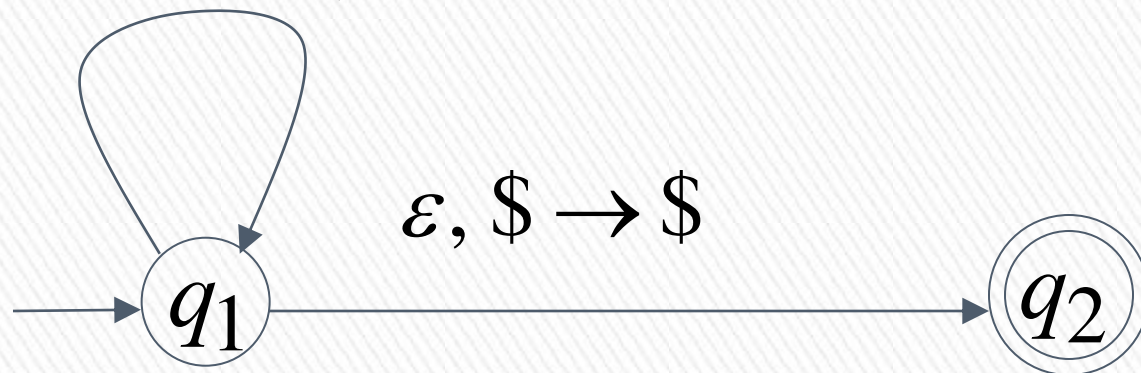
PDA M

q a u a

$a, \$ \rightarrow 0\$$ $b, \$ \rightarrow 1\$$

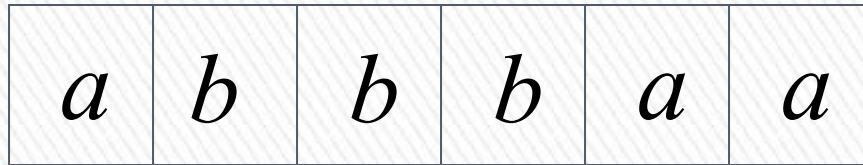
$a, 0 \rightarrow 00$ $b, 1 \rightarrow 11$

$a, 1 \rightarrow \varepsilon$ $b, 0 \rightarrow \varepsilon$

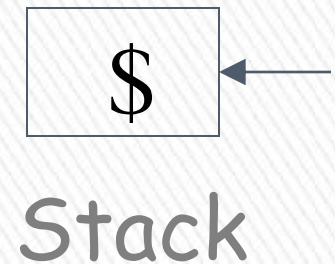


Execution Example: Time 0

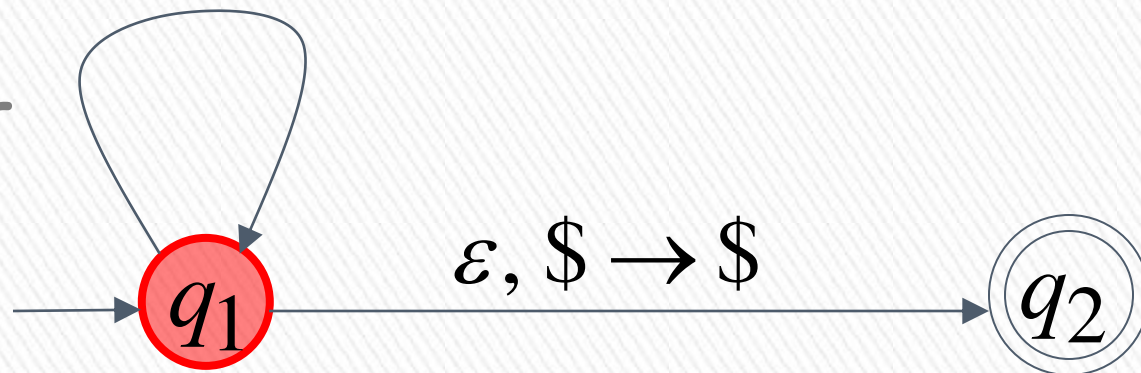
Input



$a, \$ \rightarrow a\$$ $b, \$ \rightarrow b\$$
 $a, a \rightarrow aa$ $b, b \rightarrow bb$
 $a, b \rightarrow \varepsilon$ $b, a \rightarrow \varepsilon$

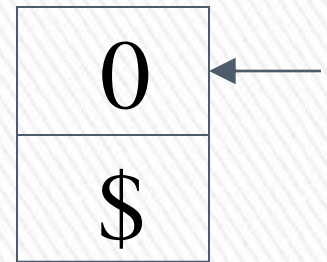
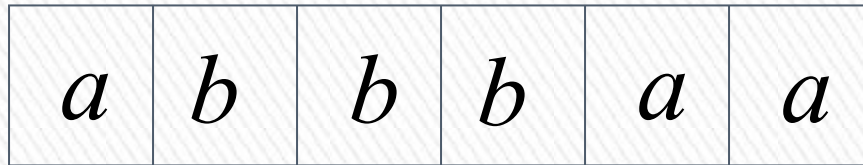


current
state



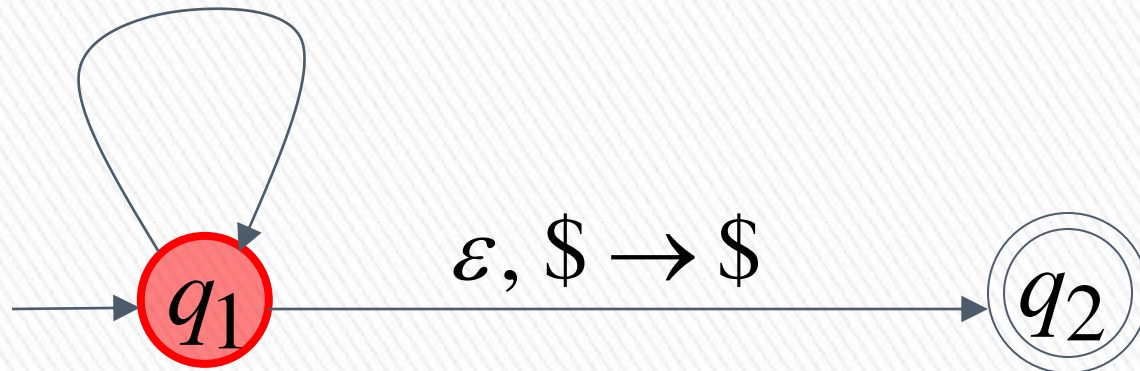
Time 1

Input



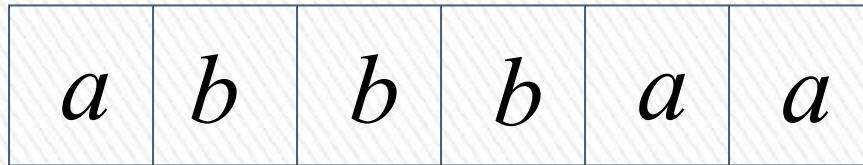
Stack

$a, \$ \rightarrow 0\$$ $b, \$ \rightarrow 1\$$
 $a, 0 \rightarrow 00$ $b, 1 \rightarrow 11$
 $a, 1 \rightarrow \varepsilon$ $b, 0 \rightarrow \varepsilon$



Time 3

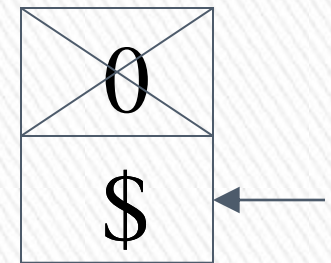
Input



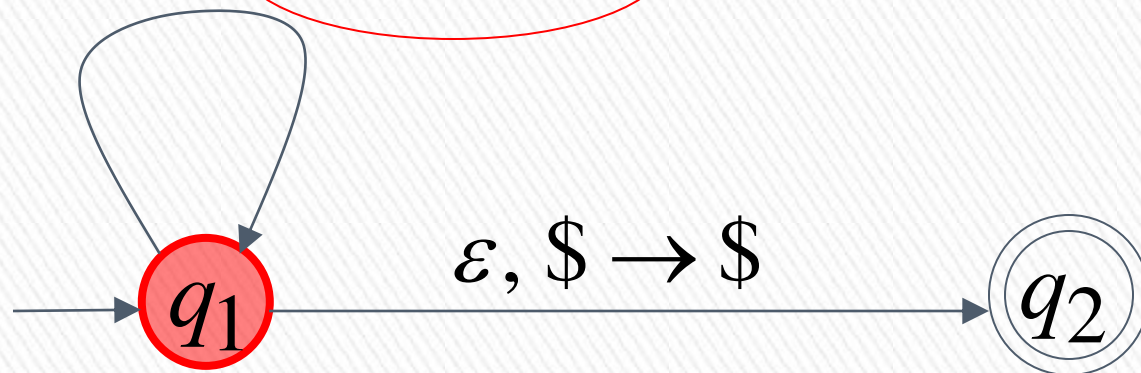
$a, \$ \rightarrow 0\$$ $b, \$ \rightarrow 1\$$

$a, 0 \rightarrow 00$ $b, 1 \rightarrow 11$

$a, 1 \rightarrow \varepsilon$ $b, 0 \rightarrow \varepsilon$

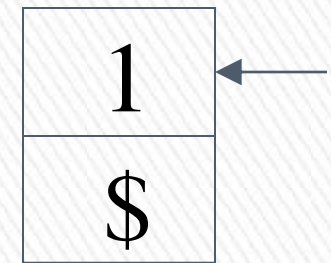
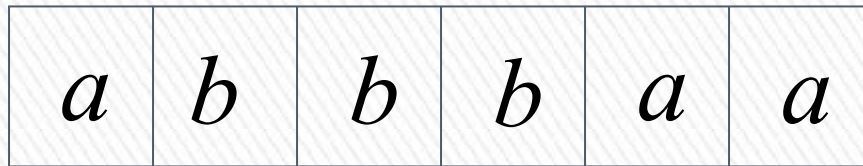


Stack



Time 4

Input

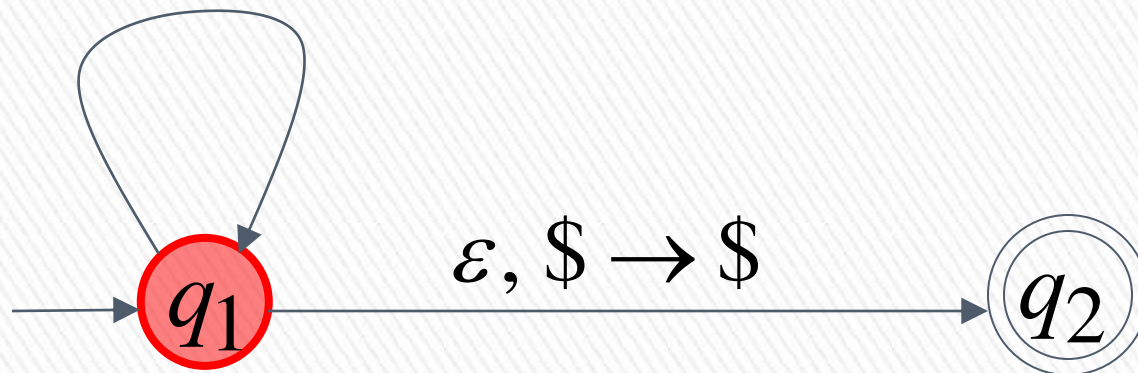


Stack

$a, \$ \rightarrow 0\$$ $b, \$ \rightarrow 1\$$

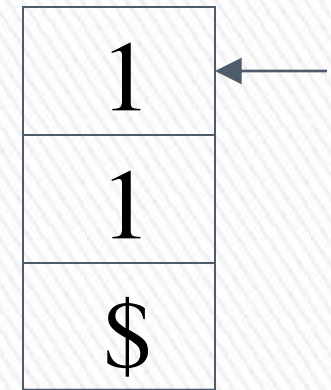
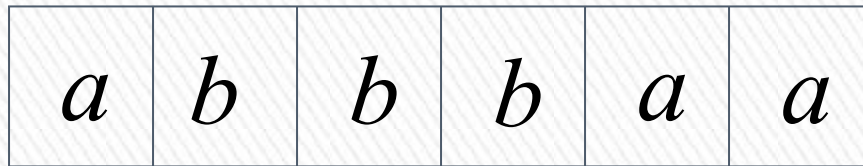
$a, 0 \rightarrow 00$ $b, 1 \rightarrow 11$

$a, 1 \rightarrow \varepsilon$ $b, 0 \rightarrow \varepsilon$



Time 5

Input

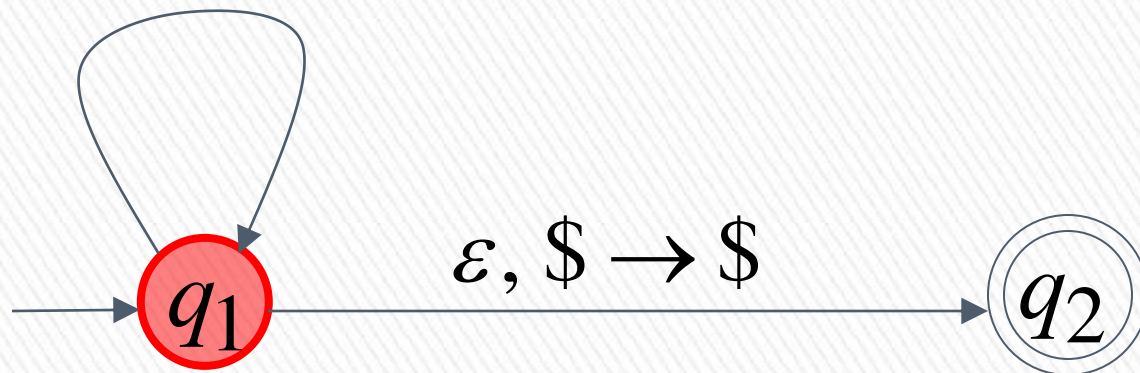


Stack

$a, \$ \rightarrow 0\$$ $b, \$ \rightarrow 1\$$

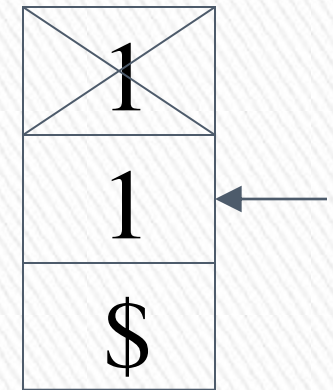
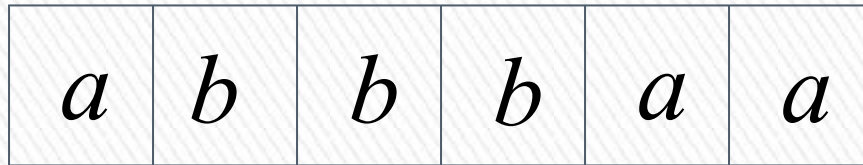
$a, 0 \rightarrow 00$ $b, 1 \rightarrow 11$

$a, 1 \rightarrow \varepsilon$ $b, 0 \rightarrow \varepsilon$



Time 6

Input

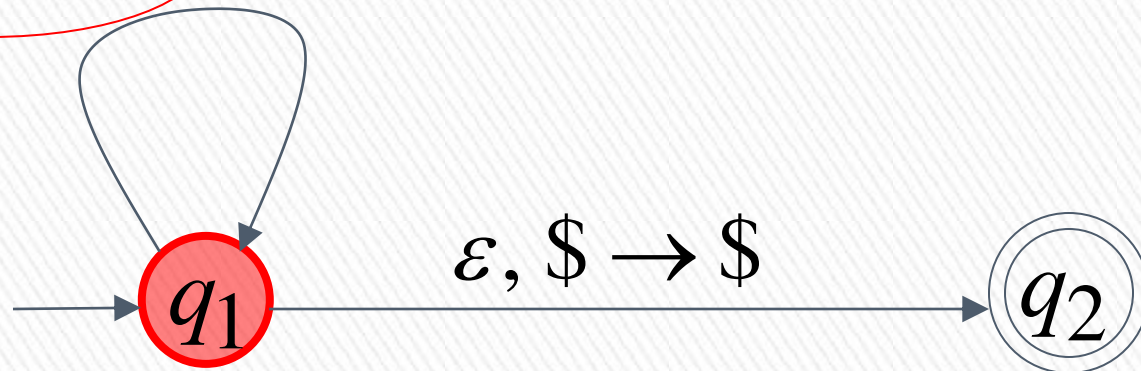


Stack

$a, \$ \rightarrow 0\$$ $b, \$ \rightarrow 1\$$

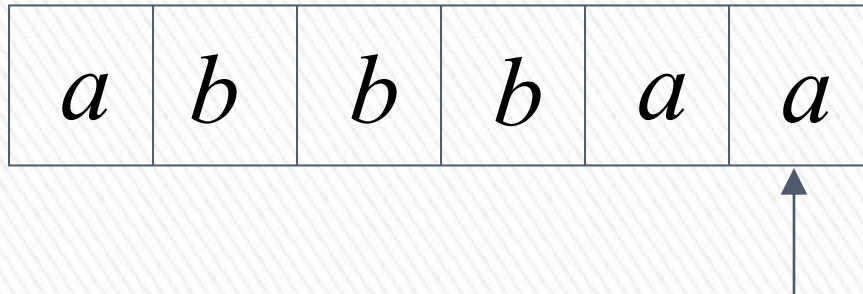
$a, 0 \rightarrow 00$ $b, 1 \rightarrow 11$

$a, 1 \rightarrow \varepsilon$ $b, 0 \rightarrow \varepsilon$



Time 7

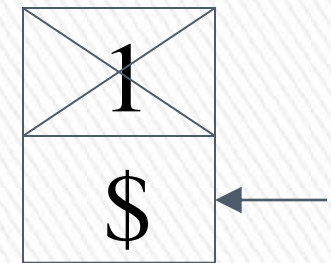
Input



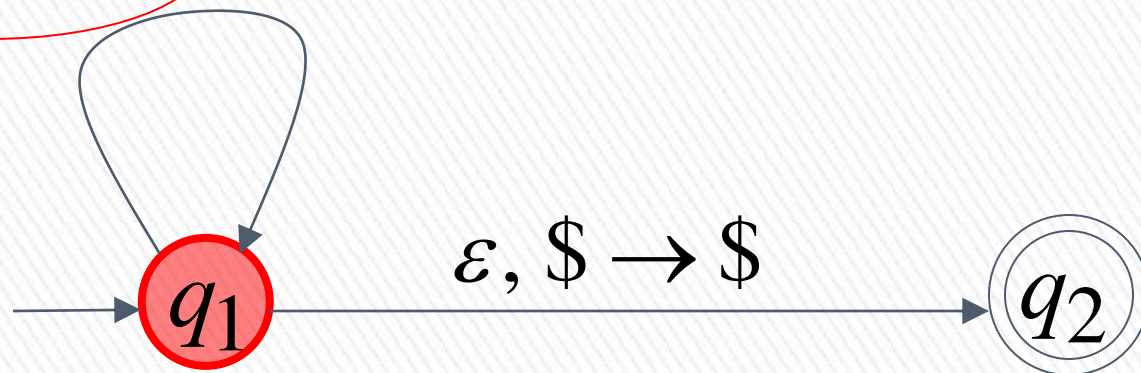
$a, \$ \rightarrow 0\$$ $b, \$ \rightarrow 1\$$

$a, 0 \rightarrow 00$ $b, 1 \rightarrow 11$

$a, 1 \rightarrow \varepsilon$ $b, 0 \rightarrow \varepsilon$

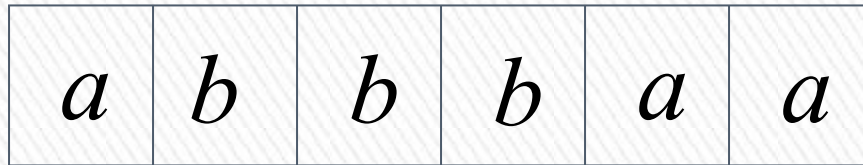


Stack



Time 8

Input



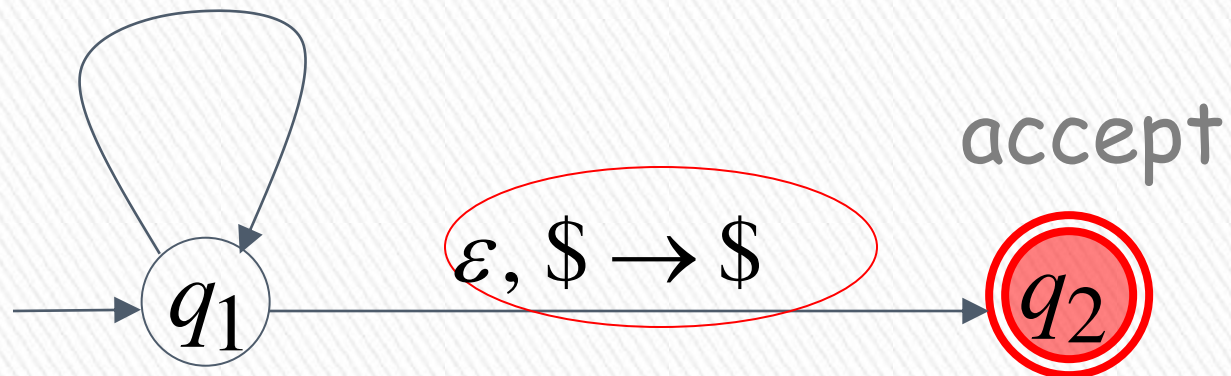
$a, \$ \rightarrow 0\$$ $b, \$ \rightarrow 1\$$

$a, 0 \rightarrow 00$ $b, 1 \rightarrow 11$

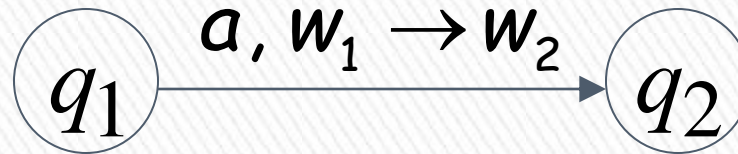
$a, 1 \rightarrow \varepsilon$ $b, 0 \rightarrow \varepsilon$



Stack



Formalities for PDAs



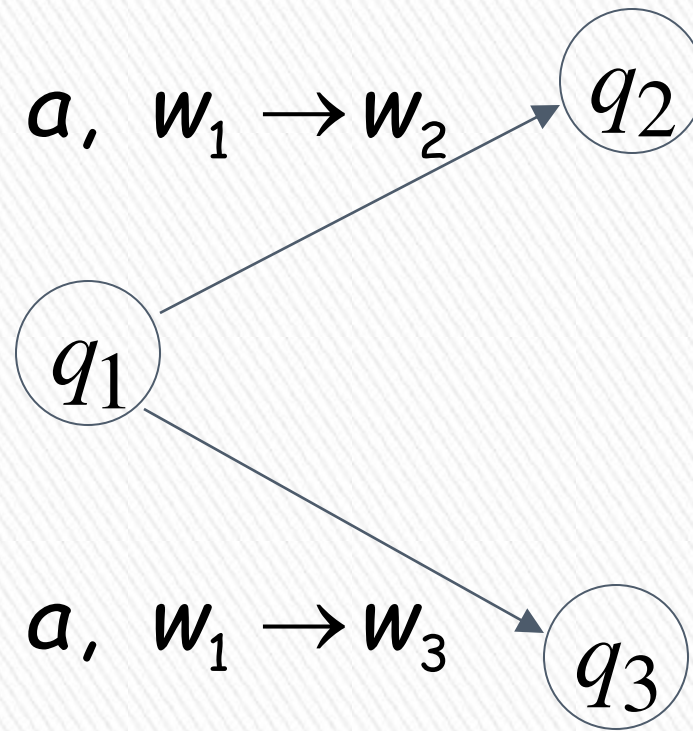
Transition function:

$$\delta(q_1, a, w_1) = \{(q_2, w_2)\}$$

Handwritten red annotations for the transition function equation:

- An arrow points from q_1 to the text "state q".
- An arrow points from a to the text "input".
- An arrow points from w_1 to the text "push-pop".
- An arrow points from q_2 to the text "state".
- An arrow points from w_2 to the text "push-pop".





Transition function:

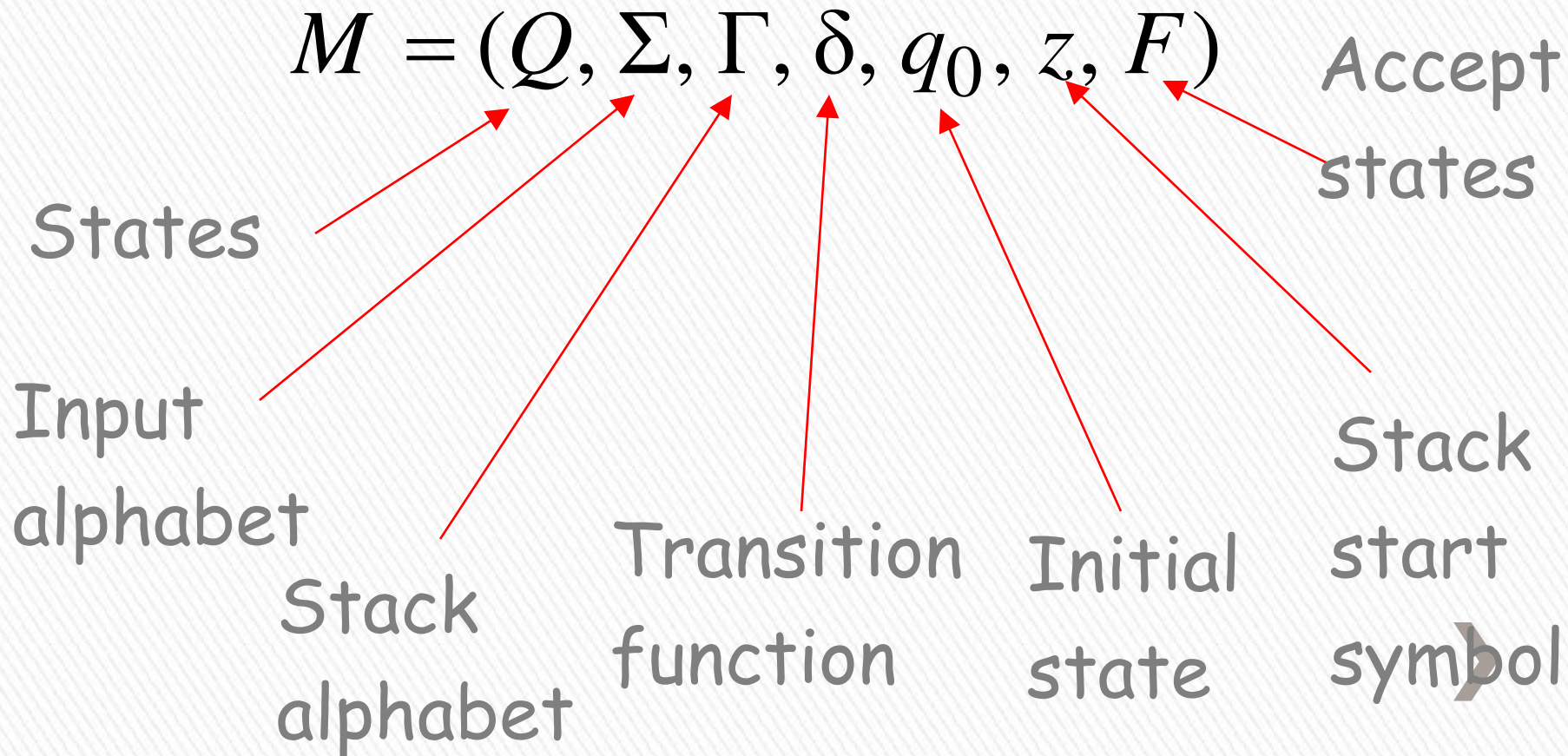
$$\delta(q_1, a, w_1) = \{(q_2, w_2), (q_3, w_3)\}$$



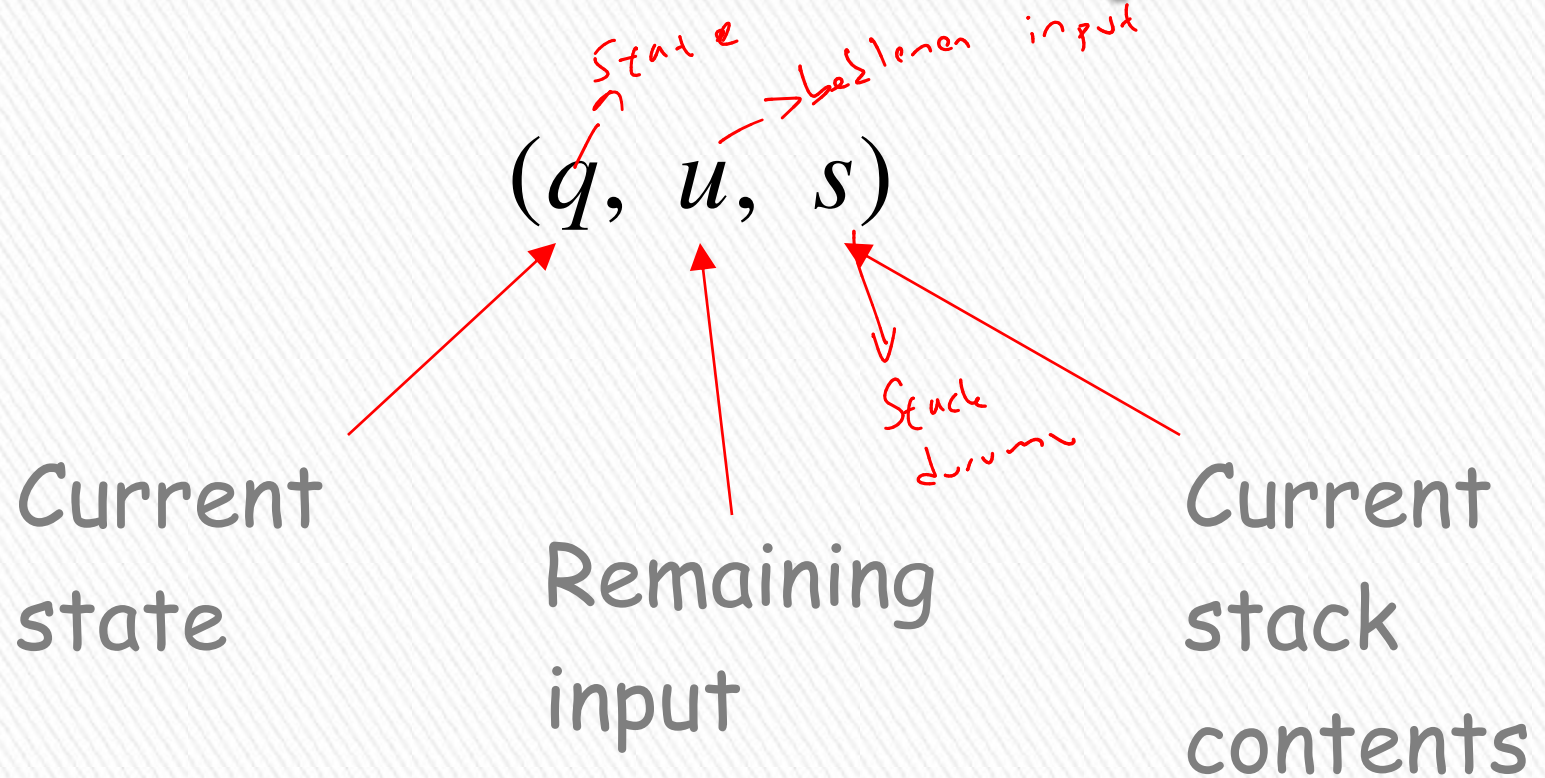
Formal Definition

Pushdown Automaton (PDA)

$$M = (Q, \Sigma, \Gamma, \delta, q_0, z, F)$$



Instantaneous Description



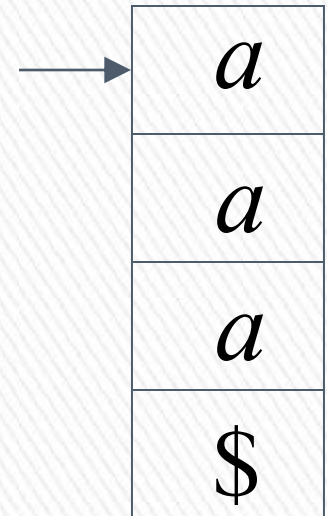
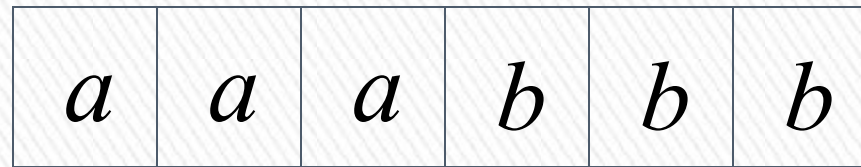
Example:

Instantaneous Description

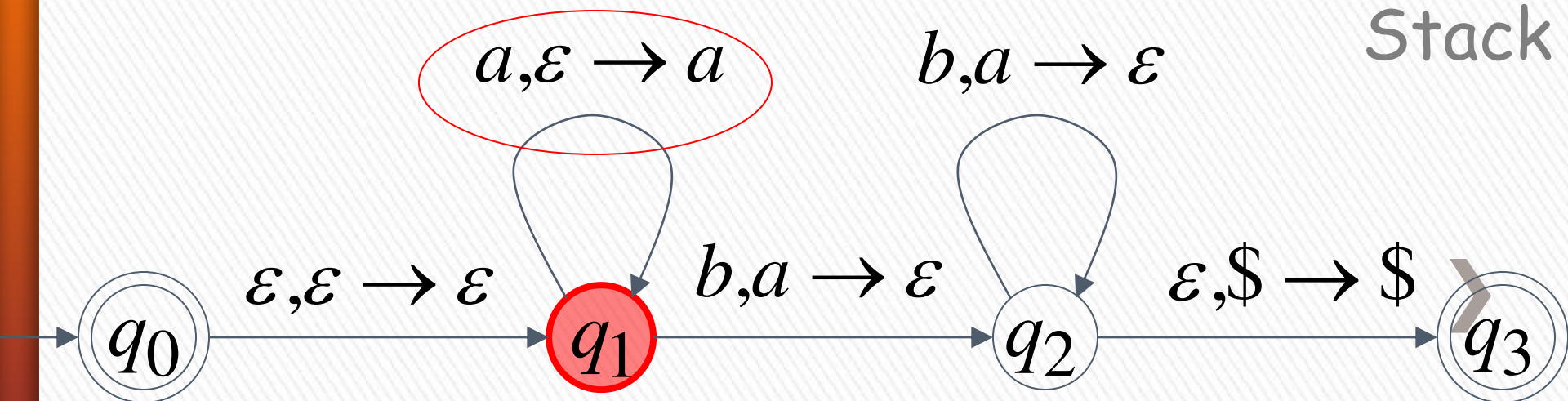
$(q_1, bbb, aaa\$)$

Time 4:

Input



Stack



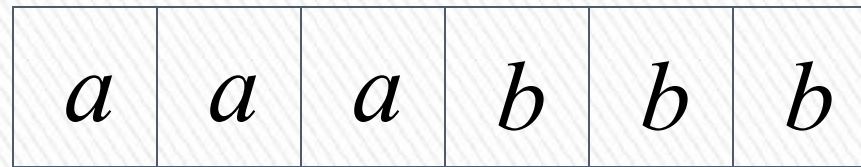
Example:

Instantaneous Description

$(q_2, bb, aa\$)$

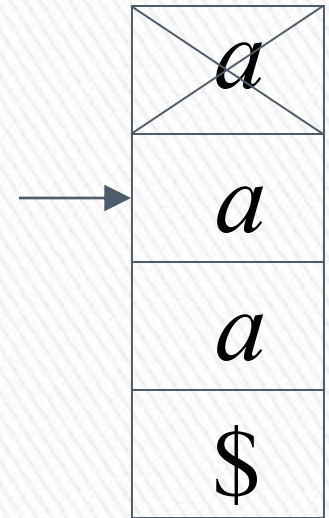
Time 5:

Input

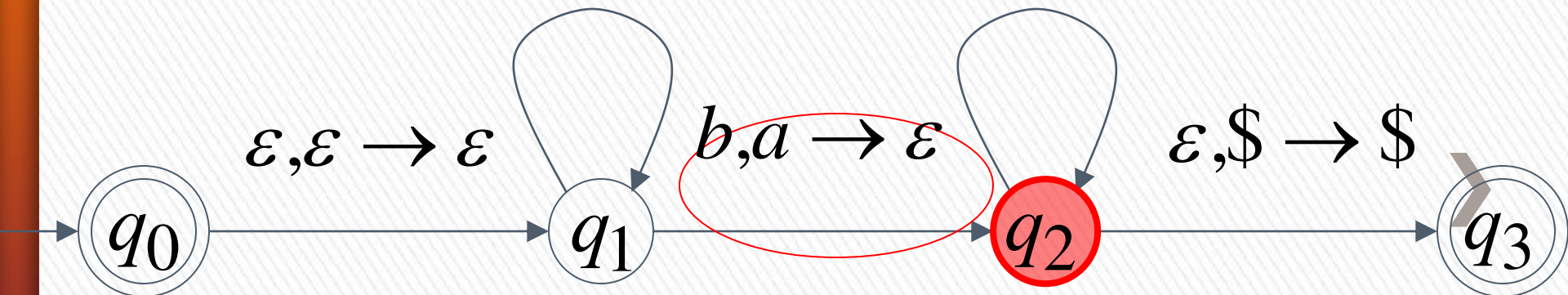


$a, \varepsilon \rightarrow a$

$b, a \rightarrow \varepsilon$



Stack



We write:

$$(q_1, bbb, aaa\$) \succ (q_2, bb, aa\$)$$

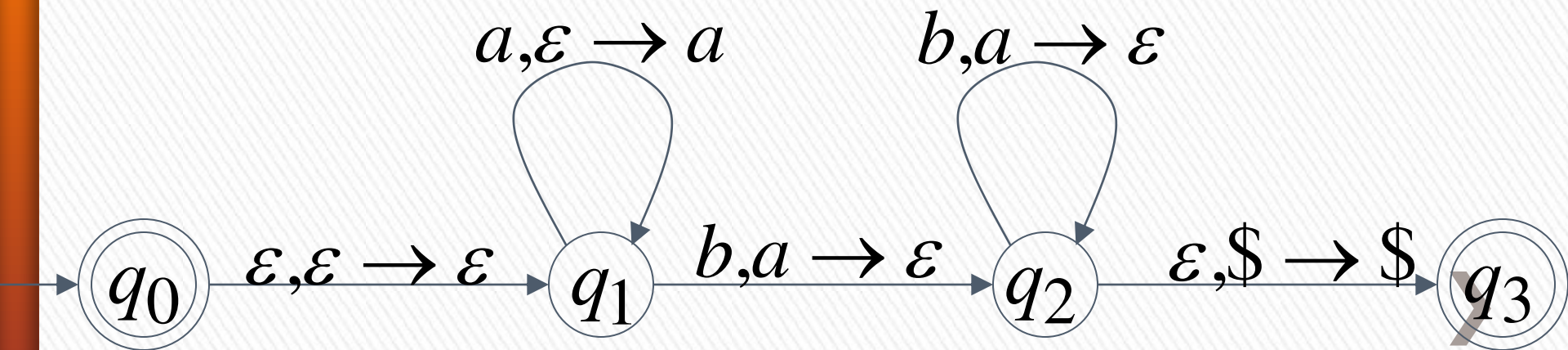
Time 4

Time 5



A computation:

$(q_0, aaabbbb, \$) \succ (q_1, aaabbbb, \$) \succ$
 $(q_1, aabbbb, a\$) \succ (q_1, abbbb, aa\$) \succ (q_1, bbbb, aaa\$) \succ$
 $(q_2, bb, aa\$) \succ (q_2, b, a\$) \succ (q_2, \varepsilon, \$) \succ (q_3, \varepsilon, \$)$



$$\begin{aligned}
 &(q_0, aaabbb, \$) \succ (q_1, aaabbb, \$) \succ \\
 &(q_1, aabbb, a\$) \succ (q_1, abbb, aa\$) \succ (q_1, bbb, aaa\$) \succ \\
 &(q_2, bb, aa\$) \succ (q_2, b, a\$) \succ (q_2, \varepsilon, \$) \succ (q_3, \varepsilon, \$)
 \end{aligned}$$

For convenience we write:

$$(q_0, aaabbb, \$) \overset{*}{\succ} (q_3, \varepsilon, \$)$$



Language of PDA $\rightarrow$ CFL *derinli* *ci:2:1:1:1*

Language $L(M)$ accepted by PDA M :

$$L(M) = \{w : (q_0, w, z) \xrightarrow{*} (q_f, \varepsilon, s)\}$$

Initial state

Accept state



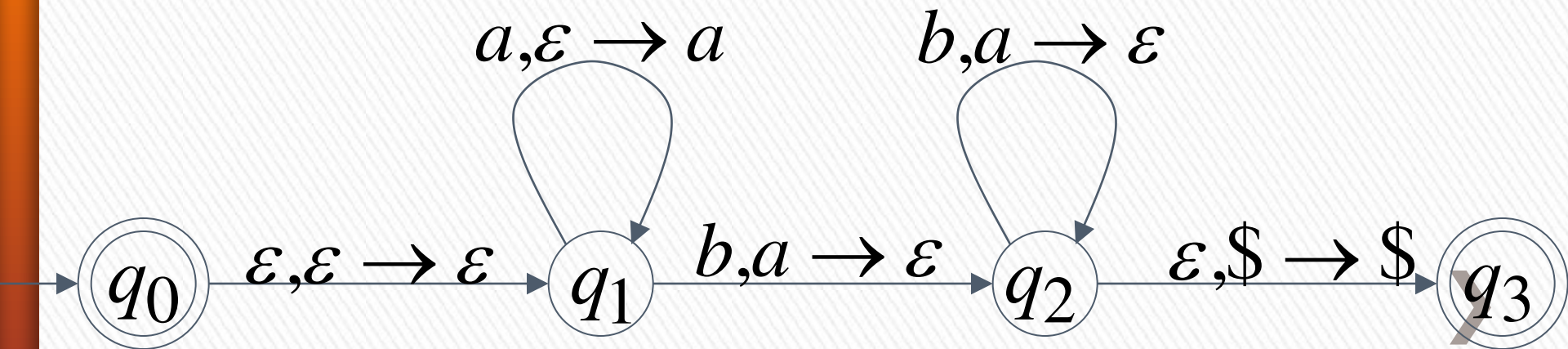
Example:

$$(q_0, aaabbbb, \$) \stackrel{*}{\succ} (q_3, \varepsilon, \$)$$



$$aaabbbb \in L(M)$$

PDA M :

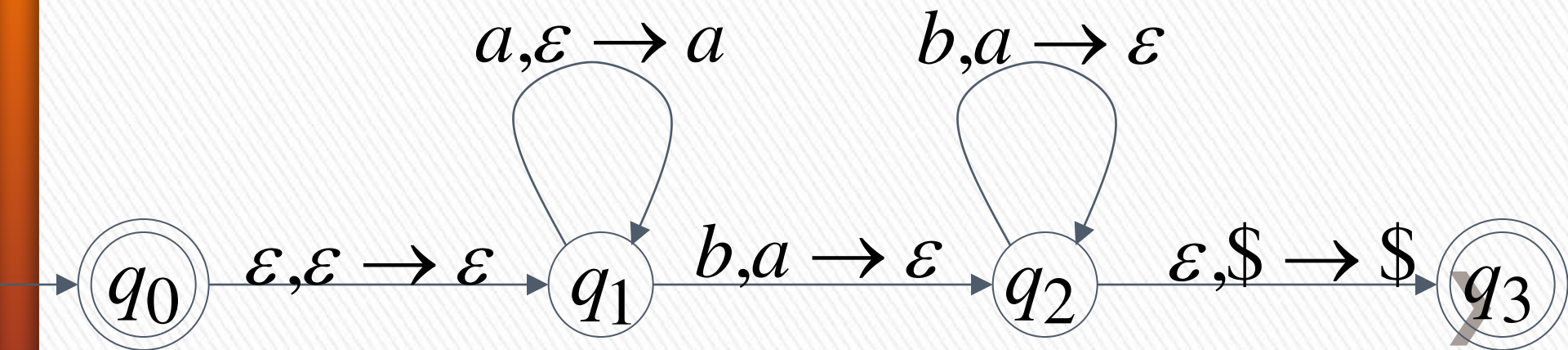


$$(q_0, a^n b^n, \$) \stackrel{*}{\succ} (q_3, \varepsilon, \$)$$



$$a^n b^n \in L(M)$$

PDA M :



Therefore:

$$L(M) = \{a^n b^n : n \geq 0\}$$

PDA M :

